Supplementary Material for "Three Routes to Failure: An Interpretable Diagnosis of Task Difficulty and Negative-Set Contamination in Genomic Sequence-Classification Benchmarks"

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https://github.com/rikhinkavuru/gb_kmer_benchmark

About this supplement

This supplement accompanies the paper in *Proc. IEEE BIBM 2026*. Section S*n* is the target of every pointer "SM S*n*" in the paper. It holds the full analyses behind the paper's robustness table (Table II of the paper) and its Methods and Results: every sweep level, dataset and design, the per-task tables, the control-motif panel and the frozen-probe figure (Figs. S5 and S6), the pre-specification record, the analyses that were packaged but not run, and the commands that reproduce everything. It is organized by concern.

- *Numbers.* Every table is written by `bibm/supplement/make_sm_tables.py` from the result files named in its caption (under `results/upgrades/` unless a path is given), and every estimate, interval and p-value quoted in the prose is a macro written by the same script; literature values, analysis settings and task counts are typed. The manifest at the end lists every input file with its SHA-256 prefix. 95% CIs are percentile bootstrap intervals (1000 resamples; the test sequence is the resampling unit unless stated). Values are rounded as shown; differences use unrounded values, and a nonzero CI or range bound that rounds to zero keeps its sign (−0.000).
- *Pre-specified and post hoc.* "Pre-specified" means that a rule or analysis was written into the analysis script (its docstring) before the analysis it governs was run; there is no external registry. "Post hoc" marks everything added later; such analyses are descriptive unless stated otherwise (Table S1).
- *Not run.* Section S9 lists analyses that are packaged or recommended but were not run. They are not results, and no number in the paper or here comes from them.
- *Splits and sources.* The *original* split is a benchmark task as distributed by Genomic Benchmarks (GB) or the Nucleotide Transformer (NT) suite. The *released equalizer* is the paper's composition equalizer (the negatives subsampled to the positives' 20-bin propensity histogram; strictness $\alpha=1$ in Section S4), and the *released equalized split* is the split it produced, distributed with the paper's code. *Published* refers only to values in the literature. Discriminability (a task's highest TF motif-only AUROC, positives scored above negatives; for a multiclass task, one class against the rest), the benchmark MCCs and the route labels of Table I are the paper's.
- *Names.* AUROC: area under the ROC curve; MCC: Matthews correlation coefficient. Short names used here ("control", "GC-balanced control" and `drosophila` all denote `drosophila_enhancers_stark`, which is GC-balanced but not composition-free: composition-only AUROC 0.683):

`cohn`	`human_enhancers_cohn`	regulatory	`human_ensembl_regulatory`
`nt_enh.`	`nt_enhancers`	`nontata`	`human_nontata_promoters`
control	`drosophila_enhancers_stark`	`enh_ensembl`	`human_enhancers_ensembl`
splice	`nt_splice_sites_all`	ocr	`human_ocr_ensembl`
`coding`	`demo_coding_vs_intergenomic_seqs`	`mouse_enh`	`dummy_mouse_enhancers_ensembl`
`human_or_worm`	`demo_human_or_worm`	`nt_enh_types`	`nt_enhancers_types`
`nt_prom_all`	`nt_promoter_all`	`nt_prom_tata`	`nt_promoter_tata`
`nt_prom_no_tata`	`nt_promoter_no_tata`		

  The *11 TF-motif tasks* (Route 1 on real tasks, Section S2) are all 15 tasks but splice and the three composition tasks (`coding`, `human_or_worm`, `mouse_enh`): the five solved promoter/regulatory tasks (regulatory, `nontata`, `nt_prom_all`, `nt_prom_no_tata`, `nt_prom_tata`) and the six failing enhancer/open-chromatin tasks (`cohn`, `nt_enh.`, `nt_enh_types`, `enh_ensembl`, ocr, control).
- *"Contamination" (Route 3)* labels a task that fails (MCC below 0.7), is not positional, and whose negatives carry a TATA-box excess of at least 10 percentage points over the positives, with a TATA motif-only AUROC below 0.5 that TATA-flag cleaning moves toward chance (the paper's TATA gate, also the rule of its diagnostic CLI;

evaluated on a seeded sample of 4,000 sequences per class and met by all three Table I contamination tasks). Such a compositional shift supplies much of the task's separability (Section S3.4). The word names a benchmark-design confound, here GC-rich positives against AT-richer negatives. It does not mean core-promoter leakage, and it does not claim that the composition difference is an artifact rather than biology.

Contents. S1 Methods and reproducibility (p. 2); S2 Route 1 on real tasks (p. 5); S3 Provenance of the negatives (p. 8); S4 Route-3 matching sensitivity and the control-motif panel (p. 10); S5 FM probes, frozen and fine-tuned (p. 18); S6 Homology re-clustering (p. 24); S7 Alternative interpretable classifiers (p. 26); S8 The GUE suite (p. 28); S9 Analyses packaged but not run (p. 32); manifest of input files (p. 34).

S1. Methods and reproducibility

S1.1 Seeds, resampling and tests

The benchmark fits use seed 42, as in the paper. The analyses of Sections S2–S8 add the following. Composition matching is re-drawn with selection seeds 42–44 (the pre-specified verdict uses seed 42), and every matching level has 10 size-matched random-negative draws (seeds 0–9 in the alternative-classifier analysis). Homology re-splits use split seeds 42–44 (seed 42 is pre-specified and gives the primary Route-1 test; label counts and per-task re-split MCCs are also given as split-seed means). Fine-tuning uses seeds 42–44. Draw dependence is measured over 20 equally valid matching draws of the released equalizer and over 16 `PYTHONHASHSEED` draws of the dinucleotide shuffle.

Per-task CIs resample test sequences. The matching decomposition uses a two-level paired bootstrap (test sequences shared by all arms, crossed with the random and matched seeds); its models are fixed, so training-set variability is not resampled. Fine-tuning contrasts resample the original test sequences jointly across arms and average over seeds. Cross-task (Route 1) statistics resample tasks. Route-1 Spearman p-values of the covariate analysis (Section S2) are exact permutation p (enumeration for $n \leq 12$; its partial-correlation p-values use t with $n-3$ df); those of the alternative-classifier analysis at $n = 11$ are Monte-Carlo permutation p; the homology analysis's Route-1 rows are reported as ρ and n because its p-values are t-approximations. Multiplicity is not corrected unless stated.

S1.2 Pre-specification record

Table S1 lists, per analysis, what was fixed before the analysis ran and what was added later. The multivariate plan is kept verbatim in `multivar_preregistration.txt` (SHA-256 of the frozen docstring `7f5ba7bd0984…`); the other plans are the docstrings of the scripts named in the table.

S1.3 Software, data and model versions

Table S2 gives the versions. All analyses except fine-tuning and tokenization are CPU-only and torch-free; fine-tuning ran on an Apple M4 GPU (PyTorch MPS) and is scored torch-free from the saved per-sequence predictions.

S1.4 DNABERT-2 token lengths of the GC-balanced control

DNABERT-2 was not run on the `drosophila` control. Table S3 measures its token lengths with the pinned tokenizer: 159/6,914 `drosophila` sequences (2.3%; 120/5,184 in training, 39/1,730 in test) exceed its 512-token context, the median is 436 tokens, and the fine-tuning code truncates at 512 tokens in any case. The omission is therefore a choice, not a context-length constraint. The control covers HyenaDNA only; the DNABERT-2 control runs are listed in Section S9 as not run.

S1.5 Reproducing the analyses

Every command runs from the repository root through the low-CPU wrapper, one heavy process at a time. Each analysis is resumable (checkpointed per unit), and its code directory has a README with outputs and measured cost. The environments:

```
PY=<torch-free venv>/bin/python   # numpy, pandas, scipy, scikit-learn, lightgbm, interpret, mappy
FT=.venv-ft/bin/python            # torch, transformers, peft, tokenizers (fine-tuning and tokenizer only)
W=bibm/lowcpu.sh                  # nice 20 and capped BLAS/OpenMP/LightGBM threads
```

Composition matching (S4):

```
M=src/upgrades/psm_sensitivity
$W $PY $M/psm_data.py; $W $PY $M/tests/test_psm_core.py; $W $PY $M/tests/test_genomic_helpers.py
$W $PY -W ignore $M/run_psm_sensitivity.py      # pre-specified run (hours; resumable)
$W $PY -W ignore $M/tests/test_reproduces_paper.py
$W $PY -W ignore $M/run_posthoc.py              # post hoc P0-P8 (resumable)
$W $PY -W ignore $M/summarize_psm.py; $W $PY -W ignore $M/tests/test_posthoc.py
$W $PY $M/make_fig_psm_sensitivity.py; $W $PY $M/make_fig_psm_sensitivity.py --caliper
$W $PY $M/make_fig_psm_sensitivity.py --nt-raw --out results/upgrades/figures/psm_sensitivity_ntraw
```

TABLE S1
PRE-SPECIFICATION RECORD. 'FIXED BEFORE' MEANS WRITTEN IN THE ANALYSIS SCRIPT (DOCSTRING) BEFORE THE ANALYSIS IT GOVERNS WAS RUN; THERE IS NO EXTERNAL REGISTRY. SOURCE: `src/upgrades/*/ (docstrings)`, `multivar_preregistration.txt`.

Analysis (code directory)	Fixed before the analysis ran *[where]*	Added later, labeled post hoc in every output
Composition matching vs. size-matched random negatives (`psm_sensitivity`)	Decision rule: criteria (i-a) size, (i-b) separation, (ii-a) overlap weighting, (ii-b) two-sided 1:1 at caliper 0.2 SD, and the verdict map; primary cell = strictness α=1 (the released equalizer rule), k-mer AUROC, retrained within-arm design, selection seed 42. Parts (1)–(6): strictness sweep, common-support trimming, 10 size-matched random seeds at every level, imbalance-robust metrics, overlap/ATT weighting with permuted-weight placebo, two-sided 1:1 caliper sweep (2–0.01 SD), positive-side matching, fixed-model and undistorted-test (train-only) designs, selection bias, label-noise checks *[run_psm_sensitivity.py (docstring)]*	P0/P0b duplicate and interval audits; P1 leak-free `nt_enhancers` test; P2 leak-free refit; P3 class-balanced training; P4/P6 between-dataset contrasts; P5 trimming selection bias; P7 decomposition CIs (extended to the fixed-model design); P8 trimming band check; dissociation statistics
Real-task covariates of learnability (`multivar`)	Covariate list and transforms; pool A (discriminability plus nine task covariates) and pool B (A plus the composition-only score, a decomposition that never enters the decision rule); analyses A0–A9; decision rule (a)–(c) *[multivar_preregistration.txt (frozen docstring)]*	P1–P13: GC-mean readout, exact permutation p ($n \leq 12$), task-identity and 31-mer locus audits, locus-independent task sets, slope transfer, orientation robustness
End-to-end fine-tuning (`fm_finetune`)	Contrasts $S{=}A{-}B$, $T{=}C{-}B$, $U{=}E{-}D$, $R{=}(E{-}0.5)/(A{-}0.5)$; rules 1–5 (rule 5: the GC-balanced control); imbalance-robust metrics; score–composition coupling; written before any v2 unit finished *[eval_ft_v2.py (docstring)]*	Share of above-chance AUROC lost; k-mer and frozen-probe baselines on identical arms; leak-free `nt_enhancers`; Q1 unit diagnostics, Q2 GC-matched effect, Q3 cCRE sensitivity, Q4 random-draw reference, Q5 within-class R^2, Q6 Bonferroni CIs
Homology re-clustering (`homology_recluster`)	MMseqs2 settings (id80 primary, id50 strict); purge vs. class-matched random-removal null; cluster-aware vs. random re-split; primary Route-1 test (the paper's discriminability vs. id80 cluster re-split MCC, 11 TF-motif tasks); label-preservation rule; X1–X10 re-runs *[run_eval.py, run_routes.py, run_extra.py (docstrings)]*	Prevalence-adjusted purge MCC; the 0.05-MCC materiality cut; split seeds 43–44 and the seed-robust call; re-pooling control for relabels; subtype-matched null; audit; containment-coverage sensitivity
Provenance of the negatives (`neg_provenance`)	Groups and background draws, features, decision rules D1–D4 *[run_neg_provenance.py (docstring)]*	Protein-coding rule background; D5; train/test split provenance and FASTA-header analysis; six-seed pooled backgrounds; ChromHMM rule counts
Alternative interpretable classifiers (`alt_model`)	Models and tuning, primary and sensitivity label rules, preservation criterion *[run_alt_model.py (docstring)]*	Mapping to the paper's Table I labels; overlap-weighting readout and contrasts; size-controlled random-negative readouts; EBM variants; uncapped refits (P1–P11)
Third suite (`gue_suite`)	Per-task pipeline with the paper's thresholds; predictions P1–P4 *[run_gue.py (docstring)]*	PH1–PH5, reference run on the paper's 15 tasks, independence-audit checks 4–7

Post hoc analyses are reported as descriptive unless the table says otherwise. Letter–number codes (A0–A9, D1–D5, P0–P13, PH1–PH5, Q1–Q6, X1–X10) are the identifiers that the analysis scripts, their READMEs and the result files use; each is described next to it here, and they carry no meaning beyond that description. The fine-tuning contrasts S, T, U and R are defined in Section S5.2; code directories are under `src/upgrades/`.

Real-task covariates (S2):

```
bash src/upgrades/multivar/run_pipeline.sh      # full pipeline (resumable); or step by step:
$W $PY src/upgrades/multivar/audit_task_overlap.py; $W $PY src/upgrades/multivar/audit_locus_overlap.py
$W $PY src/upgrades/multivar/run_multivar.py; $W $PY src/upgrades/multivar/write_interpretation.py
$W $PY src/upgrades/multivar/make_fig_multivar.py; $W $PY src/upgrades/multivar/tests/test_multivar.py
```

Provenance of the negatives (S3):

```
M=src/upgrades/neg_provenance
$W $PY $M/map_nt_enhancers.py; $W $PY $M/verify_inputs.py; $W $PY $M/tests/test_sanity.py
$W $PY $M/run_neg_provenance.py; $W $PY $M/nt_split_provenance.py; $W $PY $M/bg_seed_stability.py
$W $PY $M/chromhmm_rule_counts.py; $W $PY $M/write_sources.py; $W $PY $M/make_report.py
```

Fine-tuning (S5; training is the only GPU step):

```
M=src/upgrades/fm_finetune
$W $PY $M/prepare_splits_v2.py
nohup $M/run_queue_v3.sh >> results/upgrades/fm_finetune_v2_run.log 2>&1 &     # 54 units (MPS)
$W $PY $M/verify_preds_ft_v2.py; $W $PY $M/eval_ft_v2.py; $W $PY $M/summarize_ft_v2.py
$W $PY $M/baselines_ft_v2.py; $W $PY $M/posthoc_ft_v2.py; $W $PY $M/sensitivity_ft_v2.py
$W $PY $M/make_fig_ft_v2.py; $W $PY $M/write_interpretation_ft_v2.py
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 $W $FT $M/measure_token_lengths.py     # token lengths (Section S1.4)
```

Homology re-clustering (S6):

```
M=src/upgrades/homology_recluster
```

TABLE S2
SOFTWARE, DATA AND MODEL VERSIONS (READ FROM THE ENVIRONMENT WHEN THIS SUPPLEMENT WAS GENERATED; PINS VERIFIED IN THE FILES NAMED). SOURCE: `REPRODUCIBILITY.md`, `code-directory READMEs`.

Component	Version or pin	Recorded in
Python (analysis environment, torch-free)	3.14.5	interpreter
NumPy / pandas / SciPy	2.4.6 / 3.0.3 / 1.17.1	installed
scikit-learn / LightGBM	1.8.0 / 4.6.0	installed
interpret (EBM)	0.7.8	installed
minimap2 (mappy)	2.31	installed
MMseqs2	18-8cc5c	`mmseqs version`
PyTorch / transformers / PEFT / tokenizers (fine-tuning environment)	2.12.0 / 5.9.0 / 0.21.0 / 0.22.2	`.venv-ft`
HyenaDNA-tiny-16k-d128 revision	`d79fa37e`	`REPRODUCIBILITY.md`
DNABERT-2-117M revision	`7bce263b`	`src/upgrades/fm_finetune/README.md`
NT downstream tasks, dataset revision	`96d86d5`	`src/upgrades/neg_provenance/run_neg_provenance.py`
Genomic Benchmarks archive	Zenodo record 16605299	`src/upgrades/homology_recluster/README.md`
ENCODE SCREEN cCRE registry V4 (GRCh38)	md5 `60c14d9f`…	`src/upgrades/neg_provenance/README.md`
Motif database; gene annotation	JASPAR 2024 CORE; GENCODE v44	`src/upgrades/neg_provenance/README.md`

TABLE S3
DNABERT-2 TOKEN LENGTHS OF THE FINE-TUNING SPLITS, MEASURED WITH THE PINNED TOKENIZER (POST HOC, DESCRIPTIVE). SOURCE: `fm_finetune_v2_token_lengths.csv`.

Data	Split	*n*	Length, bp: median [IQR] (range)	Tokens: median [IQR] (range)	bp/token	>512 tokens
control	train	5,184	2,142 [2,103, 2,203] (236–3,237)	436 [425, 449] (50–664)	4.94	120 (2.3%)
	test	1,730	2,143 [2,103, 2,201] (258–3,061)	436 [426, 449] (55–633)	4.94	39 (2.3%)
	train+test	6,914	2,142 [2,103, 2,203] (236–3,237)	436 [426, 449] (50–664)	4.94	159 (2.3%)
`cohn`	train	20,843	500 [500, 500] (500–500)	100 [98, 103] (74–118)	5.10	0 (0.0%)
	test	6,948	500 [500, 500] (500–500)	100 [98, 103] (77–118)	5.10	0 (0.0%)
	train+test	27,791	500 [500, 500] (500–500)	100 [98, 103] (74–118)	5.10	0 (0.0%)
`nt_enhancers`	train	14,968	199 [199, 200] (199–200)	42 [41, 44] (30–49)	4.93	0 (0.0%)
	test	400	200 [200, 200] (200–200)	42 [41, 43] (33–48)	5.02	0 (0.0%)
	train+test	15,368	199 [199, 200] (199–200)	42 [41, 44] (30–49)	4.93	0 (0.0%)

Token counts include the [CLS] and [SEP] tokens and are taken before the fine-tuning code's truncation at 512 tokens; bp/token excludes the two special tokens. Tokenizer: `zhihan1996/DNABERT-2-117M` at revision `7bce263b` (local snapshot, offline); script `src/upgrades/fm_finetune/measure_token_lengths.py`. `drosophila` is the GC-balanced control; `cohn` and `nt_enhancers` are shown for context.

```
$W $PY $M/run_cluster.py --calibrate && $W $PY $M/run_cluster.py
$W $PY $M/run_eval.py && $W $PY $M/run_seeds.py
$W $PY $M/run_interventions.py --exp E1,E2,E3 --tiers 1,2 && $W $PY $M/run_routes.py && $W $PY $M/fix_checks.py
$W $PY $M/run_audit.py; $W $PY $M/run_e1_reconcile.py; $W $PY $M/run_linclust_calibration.py; $W $PY
       $M/run_covmode_sensitivity.py
nohup $M/run_stage_extra.sh X10 X1 X2 X5 >> external/homology_recluster/logs/stage_extra.log 2>&1 &
nohup $M/run_stage_fix1.sh >> external/homology_recluster/logs/stage_fix1.log 2>&1 &
$W $PY $M/prepare_ft_hc_splits.py && nohup $M/run_stage_ft_hc.sh >>
       external/homology_recluster/logs/stage_ft_hc.log 2>&1 &
$W $PY $M/run_labels_addendum.py && $W $PY $M/make_report_audit.py && $W $PY $M/make_report_extra.py && $W $PY
       $M/make_report.py
```

Alternative classifiers (S7):

```
M=src/upgrades/alt_model
nohup $W $M/run_all.sh > external/alt_model/logs/run_all.log 2>&1 &     # then run_post.sh, run_fix1.sh,
       run_fix2.sh
$W $PY $M/check_shuffle_hashseed.py 16; $W $PY $M/analyze_alt_model.py; $W $PY $M/make_interpretation_alt.py
$W $PY -m pytest -q $M/tests
```

GUE (S8):

```
M=src/upgrades/gue_suite
$W $PY $M/verify_gue.py; LOWCPU_THREADS=1 $W $PY -W ignore $M/gue_overlap.py
GUE_THREADS=3 $W $PY $M/run_gue.py; GUE_THREADS=3 $W $PY $M/run_gue_posthoc.py; GUE_THREADS=3 $W $PY
       $M/paper15_reference.py
GUE_THREADS=3 $W $PY -W ignore $M/run_gue_posthoc_r2.py; $W $PY -W ignore $M/rule_sensitivity.py
$M/fetch_emp_sources.sh; $W $PY -W ignore $M/emp_provenance.py
```

```
$W $PY $M/summarize_gue.py && $W $PY $M/make_interpretation_gue.py
```

This supplement:

```
$W $PY bibm/supplement/make_sm_tables.py
cd bibm/supplement && ../lowcpu.sh tectonic B290_SM.tex
```

S2. Route 1 on real tasks: discriminability, GC content and task dependence

The paper rests Route 1 on the semi-synthetic dose-response, which manipulates discriminability at arbitrary sample size. On the real tasks the relation is a consistency check only. This section gives the full analysis.

S2.1 What the real tasks show

On the 11 TF-motif tasks (all but splice and the three composition tasks; named in the About block) discriminability tracks learnability (Spearman $\rho = +0.74$, exact $p = 0.013$; leave-one-task-out (LOTO) MAE 0.14 MCC, 95% CI [0.11, 0.18]; LOTO R^2 +0.44 [−0.27, +0.70]), but GC content tracks it at least as closely ($\rho = +0.86$, $p = 0.0012$). The two are collinear ($\rho = +0.69$), and at $n = 11$ the CI of $\rho_{\mathrm{GC}} - \rho_{\mathrm{disc}}$, [−0.32, +0.53], cannot tell them apart. The class-pooled GC mean alone separates the solved promoter/regulatory tasks from the failing enhancer/open-chromatin tasks, but narrowly: the solved minimum is 0.461 (the regulatory task; 0.539 for the four promoter tasks alone) against a failing maximum of 0.440. The paper's discriminability does not separate these two clusters (Fig. S1b; `nt_enhancers_types` 0.675 against `nt_promoter_tata` 0.651).

The tasks are not independent units. They are 9 distinct sequence sets once exact copies and unions are collapsed, and the non-TATA promoter tasks of both suites and the regulatory task's promoter class share most loci (58–73%; 31-mer containment). Table S4 therefore reports the correlation on task subsets as a balanced pair. Without the four tasks with ≥5% train/test near-duplicates (8-mer Jaccard ≥0.7) discriminability gives +1.00 ($n = 7$), but GC's correlation falls to +0.71; on the 5 of those 7 with no exact sequence sharing it is +1.00 (exact $p = 0.017$) against GC +0.90 ($p = 0.083$). On the largest sets of tasks that share no loci ($n = 7$, locus link below 0.25) discriminability gives +0.71 to +0.82 ($p = 0.034$–0.088) and GC +0.93 ($p = 0.0067$). Every such set omits the regulatory task, whose pooled three-class GC mean lowers GC's correlation (without it, discriminability +0.68 and GC +0.94).

TABLE S4
Route 1 on real tasks: rank correlation of benchmark MCC with discriminability and with GC content across task subsets. Source: `multivar_posthoc_homology_clean.csv`, `multivar_posthoc_distinct_datasets.csv`, `multivar_posthoc_locus_independent.csv`, `multivar_posthoc_task_locus_links.csv`, `homology_recluster_table.csv`.

Task set	n	ρ(disc., MCC) (p)	ρ(GC, MCC) (p)	Note
All 11 TF-motif tasks	11	+0.74 (0.013)	+0.86 (0.0012)	$\rho_{\mathrm{GC}} - \rho_{\mathrm{disc}}$ [−0.32, +0.53]
Without the 4 tasks with ≥5% 8-mer-Jaccard near-duplicates	7	+1.00 (0.0004)	+0.71 (0.088)	all below the 5% 8-mer threshold
of which the 5 with no exact sequence sharing	5	+1.00 (0.017)	+0.90 (0.083)	largest locus link 6%
Leaky tasks dropped except `nt_enh_types`	8	+0.93 (0.0022)	+0.81 (0.022)	1 leaky task kept
Without `nontata` and `nt_enhancers`	9	+0.87 (0.0045)	+0.85 (0.0061)	keeps 2 leaky tasks
Largest locus-independent sets (link <0.25; 4 sets)	7	+0.71 to +0.82 (0.034–0.088)	+0.93 (0.0067)	each omits regulatory
Without the regulatory task	10	+0.68 (0.035)	+0.94 (0.0002)	
GB tasks only	6	+0.89 (0.033)	+0.89 (0.033)	
NT tasks only	5	+0.40 (0.52)	+0.87 (0.10)	3 distinct sequence sets

ρ: Spearman correlation of benchmark MCC with the paper's discriminability (disc.) or with the class-pooled GC mean (post hoc); p: exact permutation. Rows are sensitivity readouts of one correlation, not separate tests. The 11 tasks are 9 distinct sequence sets once exact copies and unions are collapsed; the non-TATA promoter tasks of both suites and the regulatory task's promoter class share most loci (58–73%; 31-mer containment). 'Near-duplicates' uses 8-mer Jaccard ≥0.7; under MMseqs2 id80, `human_ocr_ensembl` (11%) and the regulatory task (6%) also exceed 5%. Every largest locus-independent set omits the regulatory task and keeps 2 or 3 of the four leaky tasks.

S2.2 Pre-specified covariate analysis

A pre-specified multivariate analysis asked whether measured task properties explain the deviations from the discriminability fit (plan in `multivar_preregistration.txt`). Its primary validation is nested LOTO, which re-runs the whole subset search inside every fold (Table S5). The decision-rule model (pool A, selected by AICc) reaches a nested MAE of 0.143 against 0.140 for discriminability alone (ΔMAE +0.003 [−0.051, +0.066]), so no covariate set improves on discriminability, and the pre-specified verdict is that n is too small to resolve covariate effects beyond it. A four-term model with discriminability, length, within-class GC SD and train/test homology overfits: LOTO MAE 0.233 (LOTO R^2 −1.01; 0.221 with MMseqs2 homology in place of the 8-mer measure). Test-set sampling noise accounts

for 1.5% of the discriminability-only LOTO misfit (0.9% on all 15 tasks), so the misfit is between-task heterogeneity. Post hoc, that heterogeneity is the split between Table I's solved promoter/regulatory tasks and failing enhancer/open-chromatin tasks, which GC content separates narrowly and discriminability does not (Section S2.1); within the two clusters nothing is resolvable at this n.

Pool B adds the composition-only model's score. It reaches 0.117 on MCC (ΔMAE −0.023 [−0.076, +0.047], not significant) and is better on the secondary AUROC outcome (ΔMAE −0.026 [−0.046, −0.006]). Because that covariate is a weaker classifier's score on the same test set, a pool-B model decomposes the outcome rather than explaining it; it was pre-specified as a decomposition and never entered the decision rule.

Controlling for train/test homology leaves discriminability's partial correlation at +0.74 [+0.29, +0.99] with the 8-mer measure and +0.74 [+0.28, +0.99] with MMseqs2 id80 (Table S6; each covariate in Fig. S1a); at $n = 11$ these intervals are wide, and the re-split test of Section S2.3 is the stricter check. The real-task slope of MCC on discriminability, +2.61 [+1.68, +4.31], is consistent with the synthetic dose-response slope (+1.99).

TABLE S5
SELECTION-HONEST NESTED LEAVE-ONE-TASK-OUT (LOTO) COMPARISON OF COVARIATE MODELS WITH DISCRIMINABILITY ALONE. SOURCE: `multivar_nested.csv`.

Model family	Selected by	Nested MAE	Disc.-only MAE	ΔMAE [95% CI]
11 TF tasks, outcome MCC (primary)				
Pool A (disc. + 9 covariates) †	AICc	0.143	0.140	+0.003 [−0.051, +0.066]
Pool A (disc. + 9 covariates)	inner LOTO MAE	0.141	0.140	+0.001 [−0.070, +0.089]
Pool B (A + composition-only score)	AICc	0.117	0.140	−0.023 [−0.076, +0.047]
Pool B (A + composition-only score)	inner LOTO MAE	0.113	0.140	−0.027 [−0.085, +0.053]
Discriminability + 0–2 pool-A covariates	AICc	0.181	0.140	+0.041 [−0.001, +0.093]
Discriminability + 0–2 pool-A covariates	inner LOTO MAE	0.166	0.140	+0.026 [−0.028, +0.085]
11 TF tasks, outcome AUROC				
Pool A (disc. + 9 covariates)	AICc	0.073	0.062	+0.011 [−0.012, +0.036]
Pool A (disc. + 9 covariates)	inner LOTO MAE	0.085	0.062	+0.023 [−0.003, +0.054]
Pool B (A + composition-only score)	AICc	0.036	0.062	−0.026 [−0.046, −0.006]
Pool B (A + composition-only score)	inner LOTO MAE	0.034	0.062	−0.029 [−0.042, −0.015]
Discriminability + 0–2 pool-A covariates	AICc	0.080	0.062	+0.018 [+0.002, +0.039]
Discriminability + 0–2 pool-A covariates	inner LOTO MAE	0.083	0.062	+0.020 [+0.000, +0.049]
all 15 tasks, outcome MCC				
Pool A (disc. + 9 covariates)	AICc	0.267	0.182	+0.085 [+0.037, +0.131]
Pool A (disc. + 9 covariates)	inner LOTO MAE	0.277	0.182	+0.095 [+0.027, +0.178]
Pool B (A + composition-only score)	AICc	0.080	0.182	−0.101 [−0.173, −0.035]
Pool B (A + composition-only score)	inner LOTO MAE	0.069	0.182	−0.113 [−0.180, −0.051]
Discriminability + 0–2 pool-A covariates	AICc	0.286	0.182	+0.105 [+0.039, +0.183]
Discriminability + 0–2 pool-A covariates	inner LOTO MAE	0.258	0.182	+0.076 [+0.008, +0.166]

† the pre-specified decision-rule cell. ΔMAE < 0 favors the covariate model; the paired task-bootstrap CI uses the same resamples for both models. Pool B contains a performance-derived covariate (the composition-only model's score on the same test set), so a pool-B model decomposes rather than explains the outcome; it was pre-specified as a decomposition and never enters the decision rule.

TABLE S6
PARTIAL SPEARMAN CORRELATIONS ON THE 11 TF-MOTIF TASKS: DISCRIMINABILITY GIVEN EACH COVARIATE, AND THE COVARIATE GIVEN DISCRIMINABILITY. SOURCE: `multivar_partial.csv`, `multivar_posthoc_homology_models.csv`.

Covariate c	ρ(disc., MCC | c) [95% CI]	ρ(c, MCC | disc.) [95% CI]	Holm p
log mean length	+0.73 [+0.25, +0.99]	−0.15 [−0.94, +0.83]	0.14
length SD	+0.70 [−0.12, +0.99]	−0.08 [−0.84, +0.59]	0.14
GC mean	+0.39 [−0.53, +0.98]	+0.72 [+0.16, +0.98]	0.27
GC SD within class	+0.62 [−0.04, +0.99]	+0.45 [−0.53, +0.90]	0.17
GC class gap $\|d\|$	+0.70 [+0.31, +0.99]	0.00 [−0.76, +0.90]	0.14
train/test homology (8-mer Jaccard)	+0.74 [+0.29, +0.99]	−0.06 [−0.86, +0.80]	0.14
log n_{train}	+0.69 [−0.08, +0.99]	+0.34 [−0.32, +0.92]	0.14
3-class task	+0.73 [+0.34, +0.98]	−0.17 [−0.83, +0.67]	0.14
positional rescue ΔMCC	+0.83 [+0.41, +1.00]	+0.55 [−0.22, +0.95]	0.032
composition-only MCC	+0.51 [−0.39, +0.95]	+0.83 [+0.44, +1.00]	0.27
train/test homology (MMseqs2 id80; post hoc)	+0.74 [+0.28, +0.99]	–	–

Raw ρ(disc., MCC) = +0.74. CIs: task bootstrap (1000 resamples). The reverse partials and the Holm-adjusted p (disc. given c, 10 tests; p from t with $n-3$ df) are post hoc. With n=11 tasks and collinear covariates, few intervals exclude 0.

S2.3 Under homology control and under other classifiers

Under MMseqs2 homology control (Section S6) the 11-task correlation falls from +0.74 to +0.55 on the pre-specified cluster-aware re-split (not significant; t-approximation $p = 0.083$), but also to +0.57 on a random re-split (Table S7). Over split seeds 42–44 the means are +0.57 (cluster-aware) and +0.68 (random); without the two NT enhancer tasks both are +0.80, so the residual seed-mean gap comes from those two tasks, whose separately built test split any re-split pools with training data. On MCCs purged of test homologs, a sensitivity analysis, ρ is +0.88 ($n = 10$; +0.83 on the original-split MCCs of the same tasks). With MCC-tuned L1/L2 logistic regression (LR) or an EBM in place of LightGBM the correlation is not significant ($\rho = +0.27$, $p = 0.42$; $\rho = +0.47$, $p = 0.14$; $n = 11$). The real-task relation is therefore consistent with the dose-response but is not independent evidence for it.

TABLE S7
ROUTE 1 REAL-TASK CORRELATION ρ(DISCRIMINABILITY, MCC) UNDER HOMOLOGY CONTROL AND UNDER ALTERNATIVE CLASSIFIERS; CELLS GIVE ρ (n).
SOURCE: `homology_recluster_route1.csv`, `alt_model_route1.csv`.

MCC source	11 tasks	10 (no nt_enh_types)	9 (no NT enhancer tasks)
original-split MCC (Table I; reference)	+0.74 (11)	+0.83 (10)	+0.88 (9)
cluster-aware re-split, seed 42 (pre-specified primary)	+0.55 (11)	+0.60 (10)	+0.77 (9)
random re-split, seed 42 (control)	+0.57 (11)	+0.60 (10)	+0.77 (9)
cluster-aware re-split, mean of seeds 42–44	+0.57 (11)	+0.62 (10)	+0.80 (9)
random re-split, mean of seeds 42–44	+0.68 (11)	+0.73 (10)	+0.80 (9)
homologs purged (id80), raw MCC	+0.88 (10)	+0.88 (10)	+0.83 (9)
homologs purged (id80), prevalence-adjusted MCC	+0.87 (10)	+0.87 (10)	+0.82 (9)
alternative classifier: LightGBM (paper classifier)	+0.74 (11; p 0.013)	–	–
alternative classifier: MCC-tuned L1/L2 LR	+0.27 (11; p 0.42)	–	–
alternative classifier: EBM (128 k-mers)	+0.47 (11; p 0.14)	–	–

Homology rows are reported as ρ and n only: the homology analysis reports t-approximation p-values (for the primary cluster re-split, t-approximation $p = 0.083$, not significant). The purge cannot be scored for `nt_enhancers_types` (a class is eliminated), so purge rows have n=10; on the original-split MCCs of the same 10 tasks $\rho = +0.83$. Alternative-classifier rows: uncapped MCCs where available (the EBM keeps its capped MCC on the regulatory task), Monte-Carlo permutation p.

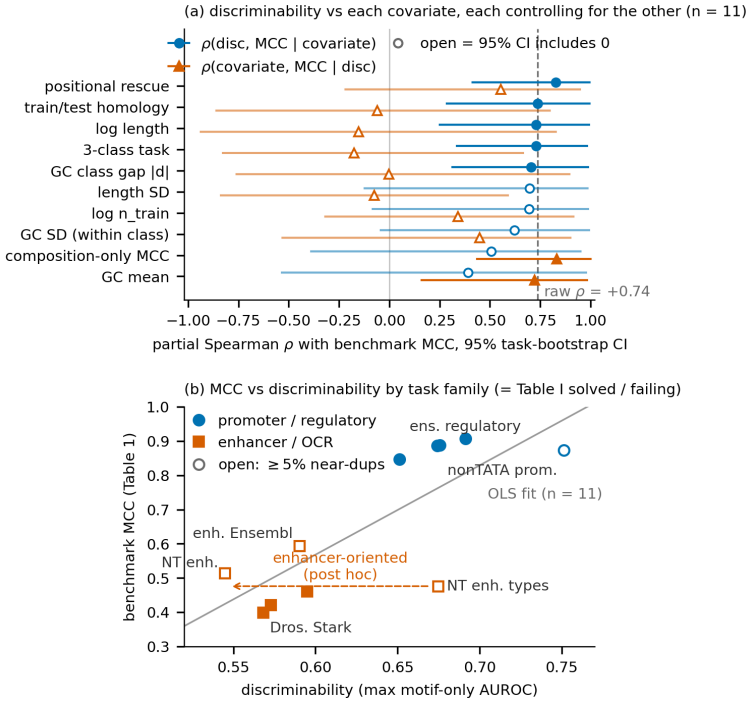


Fig. S1. Route 1 covariates on the 11 TF-motif tasks. (a) Partial Spearman correlation of benchmark MCC with discriminability given each covariate (circles) and with the covariate given discriminability (triangles); 95% task-bootstrap CIs; open symbols: CI includes 0. (b) Benchmark MCC vs. discriminability by task family (= Table I's solved/failing split; "OCR": open-chromatin); open symbols: tasks with ≥5% train/test near-duplicates by 8-mer Jaccard ≥0.7; arrow: the post hoc enhancer-oriented discriminability of `nt_enhancers_types`. The 11 tasks are 9 distinct sequence sets (Table S4). Source: `results/upgrades/figures/multivar_covariates.png` (in-figure footnote cropped).

S3. Provenance of the negatives

The paper traces the Route-3 composition gap mainly to GC-rich positives, with genomic negatives drawn from gene-distal or unmarked DNA. This section gives the evidence (Tables S8–S11).

S3.1 Where the negatives come from

`cohn` negatives are random 500-bp windows at least 15 kb from genes and outside liver H3K27ac peaks [1]. On GRCh38, 94.7% [94.3, 95.0] of them lie at least 15 kb from every GENCODE v44 protein-coding gene, against 46.2% of random windows. They are gene-distal but not gene-free: only 45.0% lie at least 15 kb from every GENCODE gene of any biotype, 33.1% lie inside a gene (32.0% in introns, mostly of non-coding genes; 1.2% in exons), 57.5% are ChromHMM state 13 in all nine lines, and 14.3% carry an enhancer state in at least one line.

Genomic `nt_enhancers` training negatives (iEnhancer-2L [2]) are ChromHMM low-signal state 13 (a nine-mark model without H3K9me3) in all nine ENCODE lines [3]: 99.8%, against 37.0% of random windows. This rule also reproduces the published candidate count (5,260,191 200-bp fragments against 5,257,994; Table S11). The test negatives (iEnhancer-EL [4]) differ: only 64.0% [57.0, 70.0] are state 13 in all nine lines, and their FASTA headers point to single state-13 segments rather than to the nine-line intersection (the construction is inferred post hoc from the headers). They are equally AT-rich (Table S10). Most of the training data are not genomic: 78.7% of the training negatives (5,889/7,484) and 68.8% of the training positives are RankGAN-synthetic sequences [5] with no locus, and only 28.2% of all `nt_enhancers` sequences (4,333/15,368, including the whole test split) are genomic.

Such DNA lies in AT-rich isochores [6]: 71.1% of `cohn` negatives and 84.0% of genomic `nt_enhancers` negatives are in L1+L2 isochores (100-kb GC below 41%), against 60.2% and 59.9% of random windows.

TATA-box rates in this section use every sequence: positives against negatives, 35.2% vs. 57.4% (+22.2 points) for `cohn` and 10.2% vs. 31.0% (+20.8) for the whole `nt_enhancers` class. The TATA gate and the motif-only scans (Section S4.8) use a seeded sample of 4,000 sequences per class, which gives 34.6% vs. 57.3% (+22.7 points) and 10.6% vs. 31.5% (+20.9).

TABLE S8
Provenance of the negatives: composition and genomic context of positives, negatives and rule-drawn or random genomic backgrounds. Source: `neg_provenance_compact_table.csv`.

Group	*n*	GC	TATA	L1+L2	≥15 kb pc	cCRE	S13 all 9	enh. ≥1
cohn: positives (H3K27ac peaks)	13,895	0.468	0.352	0.299	0.240	0.940	0.032	0.792
cohn: negatives	13,896	0.393	0.574	0.711	0.947	0.388	0.575	0.143
cohn: rule, ≥15 kb from protein-coding genes (post hoc)	10,000	0.393	0.581	0.720	1.000	0.346	0.580	0.137
cohn: rule, ≥15 kb from any GENCODE gene (pre-specified)	10,000	0.387	0.608	0.792	1.000	0.320	0.649	0.116
cohn: random genome	10,000	0.408	0.541	0.602	0.462	0.451	0.368	0.217
nt: positives (genomic, train+test)	1,584	0.489	0.119	0.232	0.226	0.814	0.000	0.999
nt: negatives (genomic, train+test)	1,684	0.374	0.356	0.840	0.685	0.189	0.955	0.015
nt: negatives, train (iEnhancer-2L benchmark)	1,484	0.375	0.357	0.846	0.695	0.187	0.998	0.000
nt: negatives, test (iEnhancer-EL independent set)	200	0.372	0.345	0.794	0.610	0.205	0.640	0.125
nt: negatives (RankGAN-synthetic, train only)	5,889	0.406	0.296	n/a	n/a	n/a	n/a	n/a
nt: rule, ChromHMM state 13 in all 9 lines (pre-specified secondary)	10,000	0.375	0.356	0.852	0.695	0.175	0.999	0.000
nt: rule, non-enhancer (not 4-7) in all 9 lines (pre-specified primary)	10,000	0.400	0.301	0.650	0.492	0.256	0.480	0.000
nt: random genome	10,000	0.410	0.285	0.599	0.462	0.338	0.370	0.214

Point estimates; the 95% bootstrap CIs (window = resampling unit) are in the source file. GC: local window GC; TATA: fraction with a TBP hit ≥1.3 bits/pos; L1+L2: 100-kb GC <41%; ≥15 kb pc: distance ≥15 kb from every GENCODE v44 protein-coding gene; cCRE: any ENCODE V4 cCRE overlap; S13 all 9: ChromHMM low-signal state 13 (a nine-mark model without H3K9me3) in all nine ENCODE lines; enh. ≥1: an enhancer state in at least one line. 'Rule' rows are backgrounds re-drawn by the documented sampling rules; which rule variant is pre-specified and which was added post hoc is stated in each row. RankGAN-synthetic sequences have no locus (n/a). TATA-box rates here use every sequence of each group. The TATA gate and the motif-only scans (Table S23) use a seeded sample of 4,000 sequences per class, which gives 34.6% vs. 57.3% (+22.7 points) for `cohn` positives vs. negatives and 10.6% vs. 31.5% (+20.9) for the whole `nt_enhancers` class; on all sequences the excess is +22.2 and +20.8 points (`nt_enhancers` whole class: 10.2% vs. 31.0%; Table S9).

S3.2 Why the negatives are AT-rich, and why that is not the whole gap

Relative to random genome, the sampling rules shift the negatives toward AT (TATA-box rate +3.3 and +7.0 points for `cohn` and genomic `nt_enhancers`). That shift is a minority of the positive–negative gap (Table S9). The negative side carries 20% [18, 23] (`cohn`) and 31% [28, 34] (genomic `nt_enhancers`) of the GC gap, and only 13.5% over the whole, mostly synthetic `nt_enhancers` class; of the TATA gap it carries 15% and 30% (11.8% over the whole class). The rest (80% and 69% of the GC gap; 87% over the whole class) is the positives' GC-richness and TATA depletion relative to random genome.

Backgrounds re-drawn by the documented rules, with unstated details inferred post hoc, reproduce the negatives' GC. For `cohn`, a rule restricted to protein-coding genes differs from the negatives by +0.0004 [−0.0009, +0.0017] in GC and +0.0108 [+0.0019, +0.0206] in TATA-box rate (six pooled background draws); for `nt_enhancers`, the state-13 rule differs by +0.0003 [−0.0035, +0.0039] and −0.0041 [−0.0293, +0.0178]. The pre-specified rule variants fare worse: the all-gene `cohn` rule over-shoots the negatives' AT shift, and the literal `nt_enhancers` rule (non-enhancer states in all nine lines) does not explain it (Table S9).

TABLE S9
HOW MUCH OF THE POSITIVE–NEGATIVE COMPOSITION GAP THE NEGATIVES' SAMPLING EXPLAINS (DECISION RULE D1). SOURCE: `neg_provenance_tests.csv`, `neg_provenance_nt_wholeclass.csv`.

Data	Metric	POS−NEG	Random−NEG	Neg.-side share [CI]	f_{rule} [CI]	D1 verdict
`cohn`	GC (window)	+0.075	+0.015	20% [18, 23]	1.36 [1.22, 1.54]	explained
`cohn`	TATA-box rate	−0.222	−0.033	15% [9, 20]	2.05 [1.54, 3.18]	explained
`cohn`	GC (100 kb)	+0.051	+0.016	31% [28, 33]	1.38 [1.30, 1.46]	explained
`cohn`	L1+L2 isochore	−0.412	−0.109	26% [24, 29]	1.75 [1.61, 1.92]	explained
genomic `nt_enh.`	GC (window)	+0.115	+0.036	31% [28, 34]	0.28 [0.21, 0.35]	not explained
genomic `nt_enh.`	TATA-box rate	−0.236	−0.070	30% [21, 37]	0.23 [0.05, 0.43]	not explained
genomic `nt_enh.`	GC (100 kb)	+0.079	+0.028	36% [34, 38]	0.23 [0.18, 0.27]	not explained
genomic `nt_enh.`	L1+L2 isochore	−0.608	−0.241	40% [37, 42]	0.21 [0.16, 0.26]	not explained
whole `nt_enh.` class*	GC (window)	+0.083	+0.011	13.5% [11, 16]	–	–
whole `nt_enh.` class*	TATA-box rate	−0.208	−0.024	11.8% [5, 17]	–	–

* post hoc; mostly RankGAN-synthetic sequences. The gap POS−NEG splits into (POS−random genome) + (random genome−NEG); the negative-side share is the second term's share: the part of the gap that comes from where the negatives were sampled. The rest is the positives' departure from random genome (their GC-richness and TATA depletion). f_{rule} = (random−rule background)/(random−NEG) for the pre-specified rule backgrounds (`cohn`: ≥15 kb from any GENCODE gene; `nt_enhancers`: literal non-enhancer states in all nine lines); $f_{\text{rule}}>1$ means the rule background over-shoots the negatives' AT shift. The verdict answers only whether that documented rule reproduces the negatives' shift from random genome. All sequences are used (for the seeded 4,000-per-class sample of the motif scan, see the note to Table S8).

S3.3 Chromosomes

The chromosome distribution of the genomic `nt_enhancers` negatives is consistent with the state-13 rule (χ^2 = 21.7, df 22, p = 0.48; homogeneity test against the rule background). That of the `cohn` negatives is not reproduced by any rule background ($p < 10^{-331}$): there are 0 chr9 negatives against 600 chr9 positives (635 expected under the rule background). This skew is unexplained and is a potential chromosome-level label shortcut.

TABLE S10
GENOMIC `nt_enhancers` NEGATIVES BY SPLIT: TRAINING (IENHANCER-2L) VS. TEST (IENHANCER-EL INDEPENDENT SET). SOURCE: `neg_provenance_nt_by_split.csv`, `neg_provenance_nt_origin.csv`.

Metric	Train negatives (n=1,484) [CI]	Test negatives (n=200) [CI]	Train positives
ChromHMM state 13 in all 9 lines	0.998 [0.995, 1.000]	0.640 [0.570, 0.700]	0.000
enhancer state in ≥1 line	0.000 [0.000, 0.000]	0.125 [0.080, 0.170]	0.999
GC (window)	0.375 [0.371, 0.378]	0.372 [0.360, 0.383]	0.491
TATA-box rate	0.357 [0.334, 0.382]	0.345 [0.275, 0.405]	0.119
L1+L2 isochore	0.846 [0.829, 0.864]	0.794 [0.734, 0.849]	0.222
any cCRE overlap	0.187 [0.168, 0.207]	0.205 [0.150, 0.260]	0.821

Synthetic share (RankGAN; no locus): 5,889 of 7,484 training negatives (78.7%), 5,146 of 7,484 training positives (68.8%); 76.6% of all 7,684 negatives and 71.8% of all 15,368 sequences; no test sequence is synthetic. The test negatives' FASTA headers point to single ChromHMM state-13 segments rather than to the nine-line intersection (inferred post hoc from the headers).

S3.4 What the provenance shows and does not show

The coordinate analysis in the paper refutes core-promoter leakage for `cohn` (TATA-flagged negatives are depleted at TSSs; Fisher $p = 3.2\times10^{-8}$); on the genomic `nt_enhancers` subset (28.2% of its sequences) the depletion is

TABLE S11
WHICH CHROMHMM DEFINITION REPRODUCES THE PUBLISHED IENHANCER-2L CANDIDATE COUNTS (POST HOC). SOURCE: `neg_provenance_chromhmm_rule_counts.csv`.

Rule (hg19 ChromHMM, nine ENCODE lines)	200-bp fragments	Published count	Ratio
state13_all9 (low signal in all nine)	5,260,191	5,257,994	1.0004
nonenhancer_all9 (not states 4-7 in all nine)	11,148,205	5,257,994	2.1202
nonenh_nonprom_all9 (not states 1-7 in all nine)	10,933,257	5,257,994	2.0794
state13or12_all9 (low signal or Repressed in all nine)	6,302,797	5,257,994	1.1987
state13to15_all9 (low signal or Repetitive/CNV in all nine)	5,287,100	5,257,994	1.0055
strongenh_all9 (states 4-5 in all nine)	742	742	1.0000
enhancer_all9 (states 4-7 in all nine)	4,222	742	5.6900
weakenh_any (states 6-7 in >=1 of nine)	2,460,484	370,517	6.6407
enhancer_any (states 4-7 in >=1 of nine)	3,004,738	370,517	8.1096
enhancer_any_not_strong_all9 (4-7 in >=1, not 4-5 in all nine)	3,003,996	370,517	8.1076

Published counts: non-enhancer candidates for the non-enhancer rules, 742 strong enhancers for the strong rule, weak-enhancer candidates for the weak rules. 'State 13 in all nine' reproduces the non-enhancer count; it does not separate state 13 from states 13–15, and no reading reproduces the weak count.

directional but underpowered ($p = 0.17$). The provenance shows that most of the composition gap is the positives' own GC-richness, with the negative-sampling choices (gene-distal windows; ChromHMM low-signal chromatin) contributing the rest. Neither is a data error, and neither separates artifactual from biological composition. Route 3's critique is one of benchmark design: on these otherwise hard tasks composition alone supplies most of the separability (composition-only AUROC 0.779 and 0.824 against k-mer 0.805 and 0.850 for `cohn` and `nt_enhancers`), whatever the composition's origin.

S4. ROUTE-3 MATCHING SENSITIVITY AND THE CONTROL-MOTIF PANEL

S4.1 Designs and the pre-specified rule

Composition equalization resamples the negatives, within each split, to the positives' histogram of a propensity score fit on the 20-dimensional mono- plus dinucleotide signature [7]. In this section *matching* means this equalization (the paper's composition-equalized intervention), and its kept negatives are the *matched* negatives; it is not the GC-matched test subset of the frozen probes (Section S5). The equalizer is *one-sided*: it removes negatives only. The released equalizer (α=1) keeps 4.5% of `cohn` negatives (max|SMD| 0.06 on the released equalized split; 0.07 on the seed-42 re-draw). We compare it with size-matched random negatives (10 draws per level) in designs that differ in what they hold fixed (Table S13):

- *within* (retrained on matched vs. random negatives, each scored on its own reduced test; unpaired). The pre-specified decision rule uses this design at strictness $\alpha = 1$ (Table S12);
- *undistorted test* (retrained on matched or random negatives, both scored on the untouched original test; paired);
- *model fixed* (the original model scored on the matched vs. random test subsets; paired);
- removal-free overlap weighting [8], retrained or with the original model, against a permuted-weight placebo;
- positive-side matching (positives subsampled to the negatives' propensity histogram; every negative kept);
- two-sided 1:1 caliper matching (both classes subsampled; the test stays balanced).

S4.2 `cohn`*: shortcut removal or sample loss?*

Random reduction to 4.5% of the negatives lowers k-mer AUROC from 0.805 to 0.765 (range 0.738–0.793), and composition matching to 0.621 [0.582, 0.658] in one matching draw (selection seed 42; seeds 42–44 give 0.613–0.637; the released equalized split 0.613). The total drop of 0.184 splits into a size effect of +0.040 [+0.022, +0.059] (22% [12, 33] of the drop) and a matched-vs-random effect of +0.144 [+0.102, +0.184] (78% [67, 88]; Table S14). The size share depends on the matching draw: over 20 equally valid draws of the released equalizer the matched AUROC spans 0.602–0.683 (median 0.636), which puts the size share at about 23% at the median draw (23.5%) and 33% at the highest draw; the mean of selection seeds 42–44 gives 22%. Because the size effect exceeds the pre-specified tolerance (0.030), the verdict is "supported, with a size component".

Test distortion and negative removal. With the test held fixed, a model trained on matched negatives scores 0.584 on the untouched original test, against 0.765 for one trained on random negatives (difference +0.181 [+0.166, +0.195]). With the model held fixed, the original model scores 0.595 on the matched test subset and 0.803 on random subsets (+0.209 [+0.169, +0.249]). Changing only the test's class ratio moves AUROC by +0.001, as expected for a prevalence-invariant metric. With no negative removed at all, matching the positives to the negatives (7.7% of positives kept,

TABLE S12
DECISION TABLE FOR COMPOSITION MATCHING AT THE RELEASED EQUALIZER'S RETENTION ($\alpha=1$; k-MER AUROC; RETRAINED WITHIN-ARM DESIGN). THE RULE AND THE cohn, ORIGINAL nt_enhancers AND CONTROL ROWS ARE PRE-SPECIFIED; THE TWO LEAK-FREE nt_enhancers ROWS APPLY THE SAME RULE POST HOC. SOURCE: psm_sensitivity_decision.csv.

Data	Original [CI]	Random (10 draws)	Matched, seed 42 [CI]	Seeds 42–44	Released	Kept	Test neg.	max|SMD|	Failed	Verdict
cohn	0.805 [0.795, 0.814]	0.765 (0.738–0.793)	0.621 [0.582, 0.658]	0.613–0.637	0.613	4.5%	228	0.07	(i-a)	supported, with a size component
nt_enhancers, original test	0.850 [0.812, 0.889]	0.807 (0.763–0.832)	0.799 [0.722, 0.871]	0.707–0.799	0.748	4.9%	36	0.30	(i-a), (i-b)	not supported
nt_enhancers, leak-free test†	0.735 [0.670, 0.790]	0.723 (0.689–0.751)	0.684 [0.587, 0.782]	0.575–0.684	0.628	4.9%	36	0.30	(i-b)	not supported
nt_enhancers, leak-free refit†	0.804 [0.760, 0.844]	0.786 (0.743–0.818)	0.706 [0.623, 0.788]	0.563–0.706	0.624	4.9%	36	0.31	(i-b)	not supported
GC-balanced control	0.763 [0.743, 0.786]	0.754 (0.729–0.775)	0.674 [0.632, 0.719]	0.650–0.674	0.661	42.9%	217	0.06	none	supported

Released: AUROC on the released equalized split. Random: mean (range) over 10 size-matched random negative draws; matched: the released equalizer rule re-drawn with selection seed 42 (the verdict uses seed 42, as pre-specified). Criteria: (i-a) size effect ≤ tolerance $\tau = \max(0.03,$ half-width of the original CI); (i-b) matched below every random seed and its upper CI below the random mean; (ii-a) overlap-weighted AUROC below every permuted-weight placebo, upper CI below the placebo mean, and a drop ≥0.5× the matching drop; (ii-b) the same for two-sided 1:1 matching at caliper 0.2 SD. SUPPORTED: all four; SUPPORTED-WITH-SIZE-COMPONENT: all but (i-a); NOT SUPPORTED: (i-b) fails or both (ii) fail. max|SMD| of the re-drawn matched set; the released equalized cohn split has 0.06. 100 of the 400 nt_enhancers test sequences, all of them positives, occur verbatim in the training split. †Post hoc: the leak-free rows remove them from every test set (leak-free test; same models) or refit every arm without their training copies and score the full test set (leak-free refit). The verdict is the same in all three nt_enhancers rows.

every negative kept) gives 0.622 [0.585, 0.659] against 0.780 for random positive subsets. Under removal-free overlap weighting the original model scores 0.609 against a placebo of 0.806, and two-sided 1:1 matching at caliper 0.2 SD gives 0.688 against 0.802.

Sweeps. Along the strictness sweep (Table S15; Fig. S2) the gap between random and matched negatives grows with balance, from +0.001 at 93% of negatives kept to +0.082 at 18% and +0.160 at 5.8%, then levels off (+0.144 at $\alpha=1$; this last step lies within the matching-draw variation above). Meanwhile the size-matched random controls fall from 0.806 to 0.765: that fall is the size component. Along the caliper sweep (Table S16; Fig. S3) random 1:1 subsets stay at 0.800–0.805, while the matched gap rises from +0.040 at a caliper of 2 SD to +0.114 at 0.2 SD, where both classes are balanced.

S4.3 What matching removes: the GC-balanced control

Matching also lowers the GC-balanced control (random 0.754, matched 0.674 [0.632, 0.719]; pre-specified verdict supported). The control has composition signal of its own (composition-only LightGBM AUROC 0.683), and in the paired designs its composition-only baseline falls as much as the k-mer model does: 0.076 against 0.072 with the model fixed, and 0.098 against 0.051 on the original test. (In the retrained within design with seed 42 the composition-only loss is smaller, 0.041 [−0.004, +0.090] against 0.081; over seeds 42–44 the two are 0.080 and 0.091.) Matching therefore removes composition-associated signal, artifactual or biological. It is also incomplete: the composition-only AUROC falls from 0.779 to 0.581 [0.544, 0.624] on the released equalized cohn split, not to 0.5.

For cohn the k-mer loss does not exceed the composition-only loss (0.144 against 0.184 retrained; 0.209 against 0.229 with the model fixed). Across the matching arms, cohn's k-mer gap correlates with the composition-only gap more than with selection depth in all three designs, but by the 0.3-margin reading rule of this post hoc summary the separation is clear only in the within design (Table S18). Under overlap weighting cohn loses a larger share of its above-chance AUROC than the control (contrast +0.353), but the composition-only model loses an equally larger share under the same weights (+0.352; difference of contrasts +0.002 [−0.070, +0.088]; Table S17). The contrast measures reliance on the composition gap, not contamination specificity.

S4.4 Selection bias and label noise

The kept cohn negatives match the positives' TATA-box rate (0.346 against 0.352; discarded 0.585) and are enriched for enhancer-like cCREs [9] (34% of kept against 24% of discarded negatives; positives 91%; Bonferroni $p = 5.3\times10^{-6}$), but not for TSS overlap after correction (0.043 against 0.024; $p = 0.77$; Table S19). If some kept negatives are unlabeled enhancers, the matched AUROC would be too low. Re-scoring without cCRE-overlapping test negatives keeps the drop (0.646 [0.596, 0.696] against 0.817, with 119 matched test negatives), and a mixture-model estimate (not a worst case) puts the AUROC cost of hidden enhancers at 0.038, 0.26 times the matched-vs-random effect (Table S20).

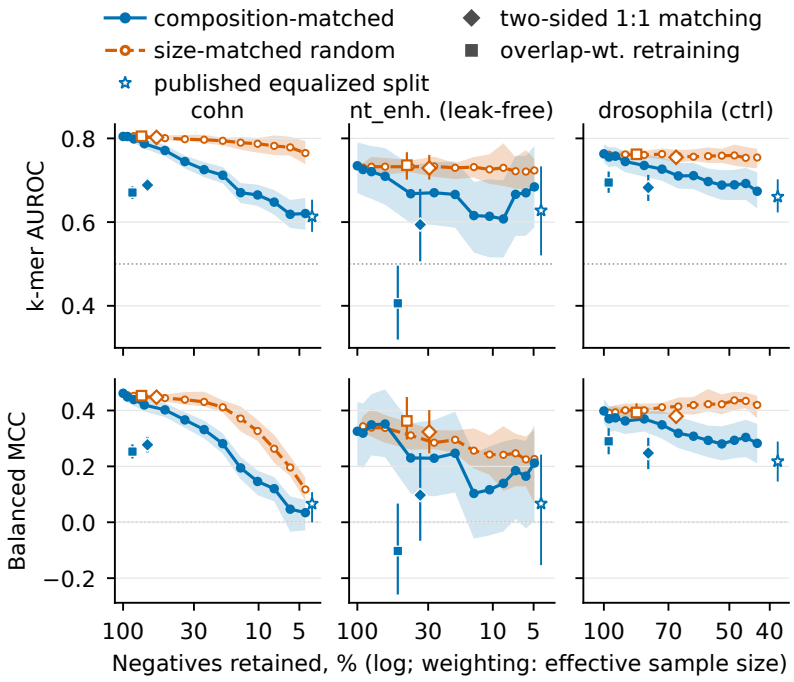


Fig. S2. Composition matching vs. size-matched random negatives across matching strictness. *k*-mer AUROC (top) and balanced MCC (bottom; class-balance-weighted, threshold = training positive fraction) against the percentage of negatives retained (log scale; for weighting, the effective sample size). Solid: composition-matched negatives (95% bootstrap CI; the last point is the released equalizer rule re-drawn with selection seed 42). Dashed: size-matched random negatives (10-draw range). Star: the released equalized split (legend: "published equalized split"). Square: model retrained with overlap weights against a permuted-weight placebo. Diamond: two-sided 1:1 caliper-0.2 matching against random 1:1. Markers are offset in x for legibility. `nt_enh.`: leak-free test (post hoc); its overlap weights over-correct, so its square lies below chance. Source: `results/upgrades/figures/psm_sensitivity.pdf`.

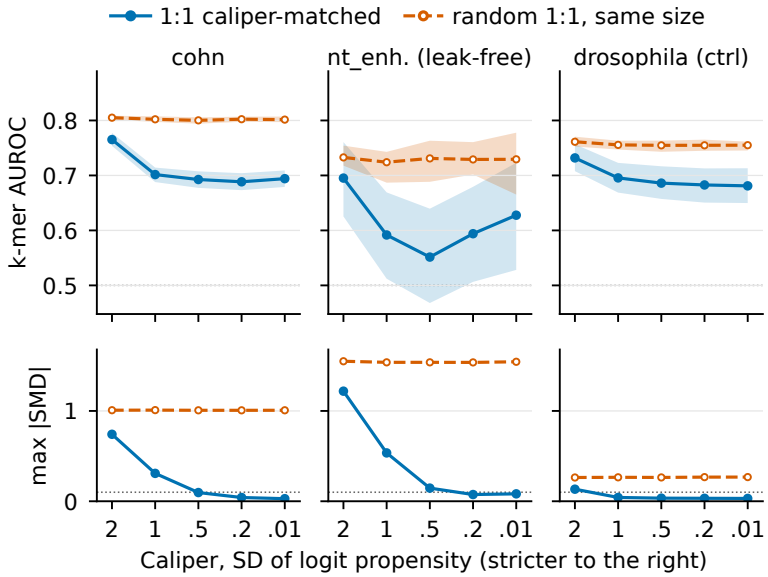


Fig. S3. Two-sided 1:1 matching across calipers from 2 to 0.01 SD of the logit propensity. Top: *k*-mer AUROC, matched (95% CI) against random 1:1 subsets of the same size (10-draw range). Bottom: max|SMD| of the matched set (solid) and of the random subsets (dashed); dotted line 0.1. Source: `results/upgrades/figures/psm_sensitivity_caliper.pdf`.

TABLE S13
DESIGNS AT $\alpha=1$ THAT CHANGE WHAT MATCHING CAN DISTORT: WHICH TEST SET IS SCORED, WHETHER THE MODEL IS RETRAINED, AND WHETHER ANY NEGATIVE IS REMOVED (k-MER AUROC; COMPOSITION-BALANCED: MATCHED WITH SELECTION SEED 42, OR OVERLAP-WEIGHTED; RANDOM / PLACEBO: MEAN OF 10 SIZE-MATCHED RANDOM DRAWS, OR OF THE PERMUTED-WEIGHT PLACEBO FOR THE WEIGHTING ROWS). SOURCE: `psm_sensitivity_decision.csv`.

Design	Pairing	Composition-balanced	Random / placebo	Difference
`cohn`				
Retrained on matched vs. random negatives, each scored on its own reduced test	unpaired (within)	0.621	0.765	+0.144
Retrained on matched vs. random negatives, scored on the untouched original test	paired (undistorted test)	0.584	0.765	+0.181
Original model scored on the matched vs. random test subsets	paired (model fixed)	0.595	0.803	+0.209
Original model on the original test, class ratio changed only	size/ratio check	–	–	+0.001
Original model, overlap-weighted original test vs. permuted-weight placebo	removal-free, paired	0.609	0.806	+0.197
Retrained with overlap weights vs. placebo	removal-free	0.671	0.805	+0.134
Positives matched to the negatives, every negative kept, vs. random positive subsets	no negative removed	0.622	0.780	+0.157
Two-sided 1:1 caliper 0.2 SD vs. random 1:1	balanced test	0.688	0.802	+0.114
`nt_enhancers`, *leak-free test*				
Retrained on matched vs. random negatives, each scored on its own reduced test	unpaired (within)	0.684	0.723	+0.039
Retrained on matched vs. random negatives, scored on the untouched original test	paired (undistorted test)	0.570	0.715	+0.145
Original model scored on the matched vs. random test subsets	paired (model fixed)	0.535	0.735	+0.200
Original model on the original test, class ratio changed only	size/ratio check	–	–	–0.001
Original model, overlap-weighted original test vs. permuted-weight placebo	removal-free, paired	0.311‡	0.738	+0.427
Retrained with overlap weights vs. placebo	removal-free	0.406‡	0.736	+0.330
Positives matched to the negatives, every negative kept, vs. random positive subsets	no negative removed§	0.464	0.746	+0.282
Two-sided 1:1 caliper 0.2 SD vs. random 1:1	balanced test	0.594	0.729	+0.135
GC-balanced control				
Retrained on matched vs. random negatives, each scored on its own reduced test	unpaired (within)	0.674	0.754	+0.081
Retrained on matched vs. random negatives, scored on the untouched original test	paired (undistorted test)	0.703	0.754	+0.051
Original model scored on the matched vs. random test subsets	paired (model fixed)	0.690	0.762	+0.072
Original model on the original test, class ratio changed only	size/ratio check	–	–	+0.001
Original model, overlap-weighted original test vs. permuted-weight placebo	removal-free, paired	0.687	0.763	+0.076
Retrained with overlap weights vs. placebo	removal-free	0.695	0.762	+0.067
Positives matched to the negatives, every negative kept, vs. random positive subsets	no negative removed	0.662	0.727	+0.065
Two-sided 1:1 caliper 0.2 SD vs. random 1:1	balanced test	0.683	0.755	+0.072

'Paired' designs hold either the model or the test set fixed, so a difference cannot come from a changed test distribution (undistorted test) or from retraining on fewer negatives (model fixed). ‡ the overlap weights over-correct (the composition-only model's weighted AUROC falls below 0.5), so only the direction is interpretable. § underpowered (17 test positives). Differences are computed from unrounded values; CIs of the within, undistorted-test and model-fixed contrasts are in Table S14.

S4.5 `nt_enhancers`

100 of the 400 `nt_enhancers` test sequences, all of them positives, occur verbatim in the training split (original AUROC 0.850; 0.735 without them), so we also score a leak-free test (post hoc) that drops them. At $\alpha=1$ the one-sided equalizer keeps 36 test negatives and cannot balance the set (max|SMD| 0.30; released equalized split 0.29). The pre-specified within-design test, on the original test, fails (matched 0.799 [0.722, 0.871] against random 0.807; matched-vs-random +0.008 [−0.074, +0.099]), and so does the same test on the leak-free test (matched 0.684 [0.587, 0.782] against random 0.723; +0.039 [−0.077, +0.145]): the verdict is "not supported" in every `nt_enhancers` variant (Table S12; Fig. S4 for the original test). On the leak-free test (post hoc), with the original model or the original test held fixed, the set loses +0.200 [+0.117, +0.294] and +0.145 [+0.055, +0.235] AUROC beyond random negatives, and the two-sided 1:1 caliper design, which does balance `nt_enhancers` (32% of each class kept; max|SMD| 0.07), gives a `cohn`-like gap (0.594 [0.506, 0.679] against 0.729, gap +0.135). In each of these designs the composition-only baseline loses at least as much (+0.236, +0.265 and +0.172). As for `cohn`, this is reliance on composition-associated signal, not contamination specificity; the control's caliper gap is +0.072.

S4.6 Why the equalized-split MCC is not used

Size-matched random negatives reproduce the equalized-split MCC collapse: for the k-mer model on `cohn`, random negatives give 0.022 (range 0.000–0.091) against 0.064 on the released equalized split, from 0.461 (share reproduced 1.11); for the composition-only baseline they reproduce 65% of its collapse (0.410 to 0.144, against 0.000 on the released equalized split; Table S21). The released split's 0.064 is also the largest of 20 equally valid matching draws

TABLE S14

DECOMPOSITION OF THE MATCHING DROP AT $\alpha=1$, WITH TWO-LEVEL PAIRED BOOTSTRAP CIS (POST HOC P7): SIZE EFFECT (ORIGINAL − RANDOM) AND MATCHED-VS-RANDOM EFFECT (RANDOM − MATCHED). SOURCE: `psm_sensitivity_decomposition_ci.csv`.

Design	Model	Size effect [CI]	Matched-vs-random [CI]	Test neg. scored
`cohn`				
within (retrained, own test)	*k*-mer	+0.040 [+0.022, +0.059]	+0.144 [+0.102, +0.184]	228
within (retrained, own test)	composition-only	+0.046 [+0.030, +0.064]	+0.184 [+0.138, +0.223]	228
within (retrained, own test), seeds 42–44	*k*-mer	+0.040 [+0.022, +0.059]	+0.141 [+0.107, +0.177]	228
within (retrained, own test), seeds 42–44	composition-only	+0.046 [+0.030, +0.064]	+0.173 [+0.133, +0.205]	228
undistorted test (retrained, original test)	*k*-mer	+0.040 [+0.034, +0.046]	+0.181 [+0.166, +0.195]	3,474
undistorted test (retrained, original test)	composition-only	+0.046 [+0.039, +0.053]	+0.216 [+0.199, +0.233]	3,474
model fixed (original model)	*k*-mer	–	+0.209 [+0.169, +0.249]	228
model fixed (original model)	composition-only	–	+0.229 [+0.187, +0.270]	228
`nt_enhancers`, *leak-free test*				
within (retrained, own test)	*k*-mer	+0.011 [−0.029, +0.052]	+0.039 [−0.077, +0.145]	36
within (retrained, own test)	composition-only	+0.018 [−0.022, +0.058]	+0.134 [−0.008, +0.273]	36
within (retrained, own test), seeds 42–44	*k*-mer	+0.011 [−0.029, +0.052]	+0.091 [−0.029, +0.217]	36
within (retrained, own test), seeds 42–44	composition-only	+0.018 [−0.022, +0.058]	+0.106 [−0.036, +0.246]	36
undistorted test (retrained, original test)	*k*-mer	+0.020 [−0.004, +0.042]	+0.145 [+0.055, +0.235]	200
undistorted test (retrained, original test)	composition-only	+0.028 [+0.006, +0.051]	+0.265 [+0.154, +0.356]	200
model fixed (original model)	*k*-mer	–	+0.200 [+0.117, +0.294]	36
model fixed (original model)	composition-only	–	+0.236 [+0.149, +0.329]	36
GC-balanced control				
within (retrained, own test)	*k*-mer	+0.009 [−0.009, +0.028]	+0.081 [+0.040, +0.123]	217
within (retrained, own test)	composition-only	+0.021 [+0.000, +0.043]	+0.041 [−0.004, +0.090]	217
within (retrained, own test), seeds 42–44	*k*-mer	+0.009 [−0.009, +0.028]	+0.091 [+0.058, +0.131]	217
within (retrained, own test), seeds 42–44	composition-only	+0.021 [+0.000, +0.043]	+0.080 [+0.031, +0.127]	217
undistorted test (retrained, original test)	*k*-mer	+0.009 [−0.000, +0.018]	+0.051 [+0.033, +0.070]	865
undistorted test (retrained, original test)	composition-only	+0.014 [+0.002, +0.028]	+0.098 [+0.075, +0.123]	865
model fixed (original model)	*k*-mer	–	+0.072 [+0.039, +0.111]	217
model fixed (original model)	composition-only	–	+0.076 [+0.039, +0.116]	217

Bootstrap: test sequences (shared by every arm) × random seeds, 1000 resamples; the models are fixed, so training-set variability is not resampled. In the model-fixed design the original model is scored on the matched vs. random test subsets, so there is no size term. The composition-only rows show how much composition signal the same selection removes; they are the reference for reading the *k*-mer rows. Within-design rows use one matching draw (seed 42) unless marked.

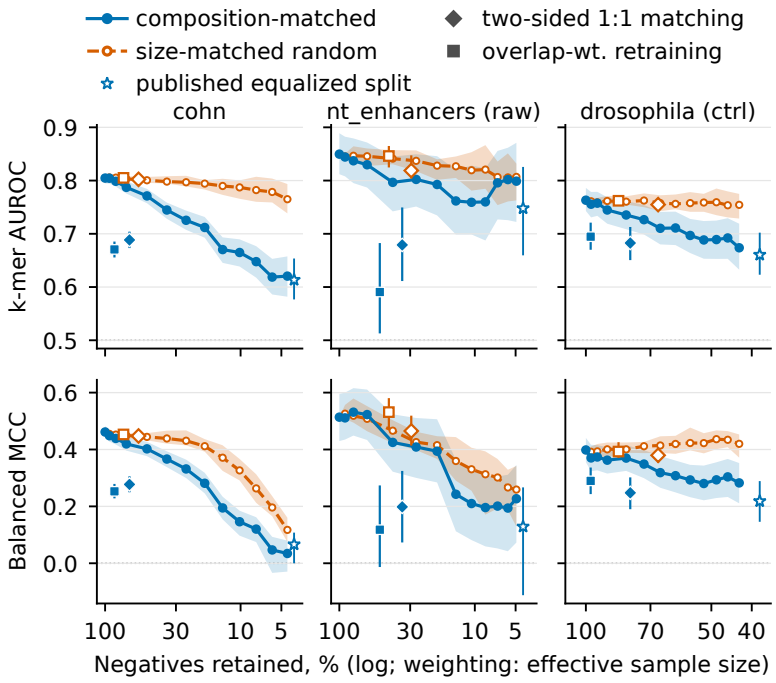


Fig. S4. As Fig. S2, with the uncorrected `nt_enhancers` test (100 of its test positives occur verbatim in training). This original test carries the pre-specified verdict (not supported); Figs. S2 and S3 use the leak-free test (post hoc). Source: `results/upgrades/figures/psm_sensitivity_ntraw.pdf`.

TABLE S15
MATCHING-STRICTNESS SWEEP: COMPOSITION-MATCHED VS. SIZE-MATCHED RANDOM NEGATIVES AT EVERY LEVEL (RETRAINED, WITHIN-ARM DESIGN; k-MER AUROC). SOURCE: `psm_sensitivity_summary.csv`.

α	Neg. kept	Test neg.	max|SMD|	Matched, seed 42 [CI]	Seeds 42–44	Random (range)	Gap	Comp.-only gap
`cohn`								
0.02	92.9%	3,255	0.99	0.804 [0.794, 0.814]	0.801–0.804	0.806 (0.802–0.808)	+0.001	+0.004
0.05	83.3%	2,957	0.96	0.799 [0.789, 0.809]	0.795–0.799	0.805 (0.804–0.809)	+0.007	+0.010
0.10	69.7%	2,526	0.91	0.787 [0.776, 0.799]	0.787–0.793	0.803 (0.800–0.808)	+0.016	+0.017
0.20	49.1%	1,852	0.81	0.771 [0.757, 0.784]	0.768–0.771	0.801 (0.796–0.807)	+0.029	+0.038
0.30	35.0%	1,376	0.71	0.745 [0.728, 0.759]	0.745–0.764	0.798 (0.794–0.808)	+0.053	+0.059
0.40	25.2%	1,031	0.61	0.725 [0.707, 0.744]	0.725–0.745	0.797 (0.789–0.809)	+0.071	+0.086
0.50	18.4%	781	0.51	0.712 [0.693, 0.732]	0.706–0.712	0.794 (0.783–0.801)	+0.082	+0.119
0.60	13.6%	597	0.41	0.670 [0.645, 0.693]	0.670–0.708	0.790 (0.773–0.803)	+0.119	+0.136
0.70	10.2%	465	0.32	0.665 [0.639, 0.690]	0.665–0.687	0.787 (0.766–0.800)	+0.122	+0.168
0.80	7.7%	364	0.24	0.648 [0.618, 0.677]	0.648–0.676	0.782 (0.760–0.807)	+0.134	+0.173
0.90	5.8%	285	0.15	0.619 [0.586, 0.653]	0.619–0.640	0.779 (0.759–0.804)	+0.160	+0.205
1.00	4.5%	228	0.07	0.621 [0.582, 0.658]	0.613–0.637	0.765 (0.738–0.793)	+0.144	+0.184
`nt_enhancers`, *leak-free test*								
0.02	90.8%	192	1.51	0.726 [0.663, 0.787]	0.726–0.740	0.730 (0.723–0.740)	+0.004	+0.010
0.05	78.7%	178	1.47	0.721 [0.656, 0.782]	0.712–0.721	0.732 (0.722–0.756)	+0.012	+0.008
0.10	62.4%	157	1.38	0.709 [0.642, 0.777]	0.704–0.723	0.732 (0.714–0.753)	+0.023	+0.023
0.20	40.4%	130	1.19	0.668 [0.597, 0.736]	0.668–0.690	0.732 (0.719–0.747)	+0.064	+0.053
0.30	27.1%	104	1.00	0.670 [0.593, 0.743]	0.670–0.690	0.733 (0.708–0.756)	+0.063	+0.072
0.40	19.0%	86	0.81	0.666 [0.587, 0.745]	0.666–0.683	0.730 (0.703–0.741)	+0.064	+0.094
0.50	13.8%	70	0.66	0.616 [0.527, 0.696]	0.616–0.659	0.731 (0.706–0.756)	+0.116	+0.107
0.60	10.6%	62	0.51	0.614 [0.516, 0.702]	0.614–0.644	0.726 (0.691–0.762)	+0.112	+0.109
0.70	8.4%	54	0.36	0.608 [0.515, 0.700]	0.608–0.625	0.730 (0.686–0.788)	+0.122	+0.137
0.80	6.8%	44	0.32	0.666 [0.567, 0.759]	0.630–0.672	0.722 (0.680–0.755)	+0.055	+0.173
0.90	5.7%	41	0.32	0.670 [0.567, 0.758]	0.620–0.670	0.721 (0.677–0.772)	+0.051	+0.131
1.00	4.9%	36	0.30	0.684 [0.587, 0.782]	0.575–0.684	0.723 (0.689–0.751)	+0.039	+0.134
GC-balanced control								
0.02	97.2%	823	0.26	0.756 [0.731, 0.779]	0.756–0.759	0.761 (0.753–0.767)	+0.006	+0.010
0.05	93.9%	782	0.25	0.758 [0.734, 0.779]	0.746–0.758	0.759 (0.751–0.767)	+0.001	+0.021
0.10	88.9%	720	0.22	0.745 [0.719, 0.769]	0.745–0.750	0.762 (0.756–0.771)	+0.017	+0.023
0.20	80.0%	615	0.20	0.735 [0.709, 0.762]	0.721–0.735	0.760 (0.755–0.771)	+0.025	+0.029
0.30	72.5%	527	0.17	0.726 [0.699, 0.754]	0.703–0.726	0.763 (0.750–0.776)	+0.036	+0.042
0.40	66.3%	459	0.14	0.710 [0.680, 0.741]	0.681–0.710	0.758 (0.743–0.772)	+0.047	+0.059
0.50	60.9%	401	0.12	0.711 [0.679, 0.742]	0.680–0.711	0.756 (0.746–0.774)	+0.045	+0.074
0.60	56.2%	349	0.10	0.697 [0.664, 0.728]	0.669–0.697	0.758 (0.741–0.778)	+0.061	+0.088
0.70	52.1%	306	0.07	0.688 [0.653, 0.724]	0.642–0.688	0.759 (0.741–0.775)	+0.070	+0.074
0.80	48.6%	271	0.06	0.689 [0.653, 0.725]	0.658–0.689	0.759 (0.735–0.785)	+0.070	+0.071
0.90	45.7%	245	0.05	0.692 [0.653, 0.729]	0.663–0.692	0.754 (0.731–0.780)	+0.061	+0.058
1.00	42.9%	217	0.06	0.674 [0.632, 0.719]	0.650–0.674	0.754 (0.729–0.775)	+0.081	+0.041

α: share of the propensity-histogram mismatch removed (α=0 keeps every negative; α=1 is the released 20-bin equalizer rule, re-drawn). Gap = random − matched (positive: matching lowers AUROC more than removing as many random negatives); comp.-only gap: the same for the composition-only LightGBM. Gaps are computed from unrounded values. Each arm is scored on its own reduced test set, so this design is unpaired; the paired designs are in Table S13. The size-matched random controls fall along this sweep (the size component), whereas along the caliper sweep (Table S16) random 1:1 subsets stay nearly flat.

(range 0.000–0.064, median 0.000). Random removal at a retention comparable to the TATA-flag cleaning (70–83% of negatives kept) lowers the `cohn` MCC by about 0.01 (0.461 to 0.451–0.453).

S4.7 Dinucleotide-shuffled negatives

Against dinucleotide-preserving shuffles of the positives [10] (no mono- or dinucleotide composition gap by construction), the negative-class TATA skew disappears (`cohn` TATA AUROC 0.551; 0.549–0.554 over 16 shuffle draws), while the positives remain separable from their own shuffles by higher-order structure (k-mer AUROC 0.95 and 0.86; Table S22). The skew is thus a by-product of the composition gap. Like matching, the shuffle does not separate artifactual from biological composition. The shuffle depends on `PYTHONHASHSEED`, and the paper's exact draw was not recovered by the 16-draw scan.

S4.8 Control-motif panel

Fig. S5 and Table S23 give the control-motif panel. Random PWMs matched to the TBP information content produce no class skew (motif-only AUROC 0.502±0.033 on `cohn`), so the scan does not manufacture one; column-scrambled TBP PWMs keep the TBP columns' AT-rich composition but not the ordered TATA pattern, and stay skewed. The skew is therefore composition-driven, not a scan artifact. The panel does not show contamination specificity.

TABLE S16

CALIPER SWEEP: TWO-SIDED 1:1 NEAREST-NEIGHBOUR MATCHING ON THE LOGIT PROPENSITY VS. RANDOM 1:1 SUBSETS OF THE SAME SIZE (*k*-MER AUROC; PRE-SPECIFIED PART 4B). SOURCE: `psm_sensitivity_caliper_sweep.csv`.

Caliper (SD)	Kept/class	Test pos./neg.	SMD matched	SMD random	Matched [CI]	Random 1:1 (range)	Gap	Comp.-only gap
`cohn`								
2	89%	3,125/3,125	0.74	1.01	0.765 [0.754, 0.777]	0.805 (0.801–0.809)	+0.040	+0.047
1	73%	2,574/2,574	0.31	1.01	0.702 [0.688, 0.714]	0.802 (0.797–0.808)	+0.101	+0.124
0.5	65%	2,279/2,279	0.10	1.01	0.693 [0.677, 0.707]	0.800 (0.794–0.806)	+0.108	+0.171
0.2	61%	2,158/2,158	0.04	1.01	0.688 [0.673, 0.704]	0.802 (0.798–0.807)	+0.114	+0.170
0.01	60%	2,088/2,088	0.03	1.01	0.694 [0.679, 0.709]	0.802 (0.797–0.807)	+0.107	+0.170
`nt_enhancers`, *leak-free test*								
2	82%	91/184	1.22	1.55	0.695 [0.625, 0.761]	0.733 (0.718–0.754)	+0.038	+0.050
1	53%	74/145	0.54	1.54	0.592 [0.512, 0.669]	0.724 (0.687–0.743)	+0.132	+0.133
0.5	38%	67/122	0.15	1.54	0.552 [0.468, 0.640]	0.731 (0.688–0.763)	+0.179	+0.181
0.2	32%	62/111	0.07	1.54	0.594 [0.506, 0.679]	0.729 (0.702–0.761)	+0.135	+0.172
0.01	30%	44/81	0.08	1.54	0.628 [0.528, 0.723]	0.729 (0.665–0.778)	+0.102	+0.155
GC-balanced control								
2	91%	783/783	0.13	0.26	0.732 [0.708, 0.757]	0.761 (0.753–0.770)	+0.029	+0.060
1	80%	677/677	0.04	0.27	0.696 [0.669, 0.723]	0.756 (0.747–0.763)	+0.060	+0.097
0.5	75%	627/627	0.03	0.26	0.686 [0.657, 0.717]	0.755 (0.743–0.763)	+0.069	+0.112
0.2	72%	607/607	0.03	0.27	0.683 [0.651, 0.713]	0.755 (0.745–0.765)	+0.072	+0.118
0.01	69%	546/546	0.03	0.27	0.681 [0.650, 0.713]	0.755 (0.745–0.761)	+0.074	+0.110

Both classes are subsampled to the same fraction, so the test stays balanced. Kept/class: retained fraction of negatives (for the leak-free `nt_enhancers` test the positive fraction differs by at most about one point, because the leak-free removal touches positives only). SMD: max|SMD| over the 20 composition features.

TABLE S17

BETWEEN-DATASET CONTRAST UNDER REMOVAL-FREE OVERLAP WEIGHTING: SHARE OF ABOVE-CHANCE AUROC LOST BY THE *k*-MER MODEL AND BY THE COMPOSITION-ONLY MODEL UNDER THE SAME WEIGHTS (POST HOC P4/P6). SOURCE: `psm_sensitivity_between_dataset.csv`.

Data	*k*-mer share lost [CI]	Comp.-only share lost [CI]	*k*-mer/comp.-only [CI]
original model, overlap-weighted test			
`cohn`	0.644 [0.606, 0.685]	0.806 [0.760, 0.855]	0.80 [0.77, 0.83]
GC-balanced control	0.290 [0.248, 0.336]	0.454 [0.376, 0.545]	0.64 [0.52, 0.75]
`cohn` − control	+0.353 [+0.293, +0.413]	+0.352 [+0.247, +0.442]	diff. of contrasts +0.002 [−0.070, +0.088]
retrained with overlap weights			
`cohn`	0.441 [0.403, 0.477]	0.622 [0.577, 0.665]	0.71 [0.64, 0.78]
GC-balanced control	0.257 [0.209, 0.310]	0.364 [0.278, 0.460]	0.71 [0.54, 0.95]
`cohn` − control	+0.183 [+0.119, +0.244]	+0.258 [+0.153, +0.356]	diff. of contrasts −0.074 [−0.186, +0.031]

The *k*-mer contrast between `cohn` and the GC-balanced control is matched by the composition-only contrast under the same weights (difference of contrasts near 0), so it measures reliance on the composition gap, not contamination specificity. `nt_enhancers` is omitted: the weights over-correct there (weighted AUROC below 0.5), so its shares (>1) are not interpretable.

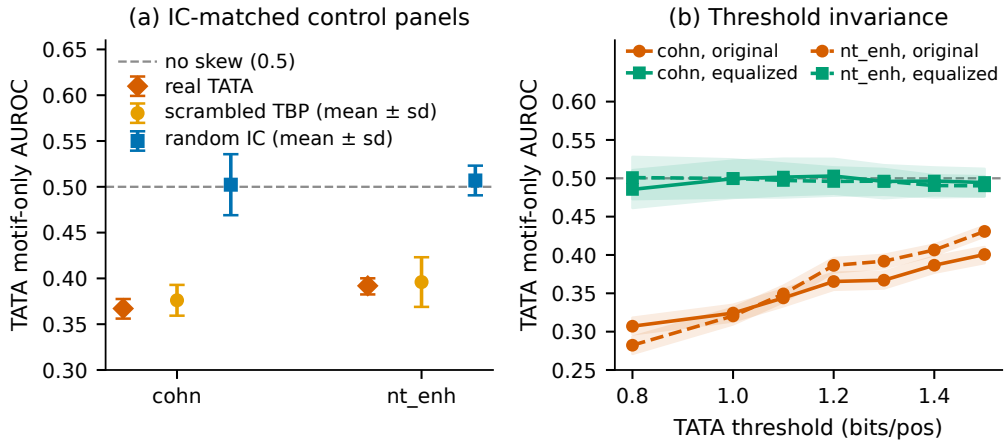


Fig. S5. Control-motif panel (data and sources as in Table S23; seed 42; at most 4,000 sequences per class). (a) TATA motif-only AUROC (positives > negatives) of the real TBP PSSM (95% CI), 20 column-scrambled TBP PWMs (composition kept, order destroyed) and 10 random PWMs matched to TBP's information content (both mean ± SD), on `cohn` and `nt_enhancers` (`nt_enh`); dashed line: no skew. (b) The same AUROC across 0.8–1.5 bits/pos on the original and the released equalized splits (95% bootstrap bands). Re-plot of `results/upgrades/figures/route3_specificity.pdf` (same code and data) with the legend of (b) moved.

TABLE S18
ACROSS MATCHING ARMS, DOES THE k-MER GAP TRACK THE COMPOSITION SIGNAL REMOVED OR THE SELECTION DEPTH? (POST HOC, DESCRIPTIVE). SOURCE: `psm_sensitivity_dissociation_stats.csv`.

Data	Design	Arms	ρ w/ comp. gap	ρ w/ depth	OLS comp.	OLS depth	Reading (0.3-margin rule)
`cohn`	within (retrained, own test)	24	+0.96	+0.39	+0.045	+0.016	gap tracks composition-signal removal more than selection depth
	model fixed (original model)	24	+0.98	+0.71	+0.061	+0.005	no clear separation of composition-signal removal and selection depth
	undistorted test (retrained, original test)	24	+0.92	+0.65	+0.070	+0.024	no clear separation of composition-signal removal and selection depth
`nt_enhancers`, leak-free test	within (retrained, own test)	22	+0.81	+0.18	+0.040	−0.007	gap tracks composition-signal removal more than selection depth
	model fixed (original model)	22	+0.99	+0.63	+0.067	+0.001	gap tracks composition-signal removal more than selection depth
	undistorted test (retrained, original test)	24	+0.86	+0.63	+0.085	−0.022	no clear separation of composition-signal removal and selection depth
GC-balanced control	within (retrained, own test)	24	+0.07	+0.86	+0.016	+0.038	gap tracks selection depth (-log retention) more than composition-signal removal
	model fixed (original model)	24	+0.99	+0.76	+0.040	−0.001	no clear separation of composition-signal removal and selection depth
	undistorted test (retrained, original test)	24	+0.97	+0.81	+0.025	+0.019	no clear separation of composition-signal removal and selection depth

Spearman correlation of the k-mer gap with the composition-only gap and with selection depth (− log retained fraction) over the non-underpowered strictness, trimming and caliper arms; OLS: partial coefficients per SD. Arms overlap and the random controls are nested, so no p-values are given. For `cohn` the correlation with the composition gap is the larger of the two in all three designs, but by the 0.3-margin rule the separation is clear only in the within design.

TABLE S19
SELECTION BIAS OF COMPOSITION MATCHING ON `cohn` (α=1, SEED 42): FEATURES OUTSIDE THE PROPENSITY MODEL IN KEPT VS. DISCARDED NEGATIVES. SOURCE: `psm_sensitivity_selection_bias.csv`.

Feature	Kept neg. (n=627)	Discarded (13,269)	Random, same size	Positives	Bonferroni p
TATA-box hit (TBP ≥1.3 bits/pos)	0.346	0.585	0.576	0.352	1.4×10^{-29}
ENCODE enhancer-like cCRE (ELS) overlap	0.337	0.236	0.239	0.907	5.3×10^{-6}
any ENCODE cCRE overlap	0.512	0.382	0.391	0.940	2.3×10^{-8}
promoter-like cCRE (PLS) overlap	0.011	0.003	0.003	0.067	0.52
GENCODE v44 TSS ±500 bp overlap	0.043	0.024	0.026	0.156	0.77
RepeatMasker coverage	0.539	0.552	0.555	0.286	1.0

Fisher or Mann–Whitney test of kept vs. discarded negatives, Bonferroni-adjusted over 158 tests (part 6, pre-specified). The kept negatives match the positives' TATA rate and are enriched for enhancer-like cCREs, the expected signature if some are unlabeled enhancers; Table S20 checks what that could do to the AUROC.

TABLE S20
LABEL-NOISE CHECKS ON `cohn`: DOES MATCHING LOWER AUROC ONLY BECAUSE KEPT NEGATIVES ARE UNLABELED ENHANCERS? SOURCE: `psm_sensitivity_decision.csv`.

Check	Matched [CI]	Random	Matched test neg.
cCRE-overlapping test negatives removed, same models (pre-specified design c)	0.646 [0.596, 0.696]	0.817	119
Two-sided 1:1 on the cCRE-purged pool (design b; balanced)	0.755 [0.738, 0.772]	0.862	–
Re-matching on the cCRE-purged pool (design a; not balanced)	0.686 [0.610, 0.762]	0.779	–

Design (a) keeps 0.9% of negatives and is not composition-balanced (max|SMD| 0.17); design (b) is balanced (max|SMD| 0.08). A mixture-model *estimate* (not a worst case) of the AUROC cost of hidden enhancers among the kept negatives is 0.038, 0.26× the matched-vs-random effect.

TABLE S21
WHY THE EQUALIZED-SPLIT MCC IS NOT USED: SIZE-MATCHED RANDOM NEGATIVES REPRODUCE MOST OR ALL OF ITS COLLAPSE (MCC AT THRESHOLD 0.5). SOURCE: `psm_sensitivity_decision.csv`.

	k-mer LightGBM				composition-only LightGBM			
Data	Original	Random (range)	Released split	Share	Original	Random	Released split	Share
`cohn`	0.461	0.022 (0.000–0.091)	0.064	1.11	0.410	0.144	0.000	0.65
`nt_enhancers`, leak-free test	0.311	0.191 (0.065–0.290)	0.000	0.39	0.310	0.131	0.060	0.72
GC-balanced control	0.398	0.425 (0.398–0.477)	0.177	–0.12	0.253	0.257	0.049	–0.02

Share = (original − random)/(original − released equalized split), the share of the collapse that size-matched random negatives reproduce. On the `cohn` test set, which is about 94% positive at this retention, a model trained on about 96% positives predicts almost everything positive, so MCC at 0.5 mostly measures the size/imbalance change. Over the 20 equally valid matching draws of the released equalizer on `cohn`, the equalized *k*-mer MCC ranges 0.000–0.064 (median 0.000; `homology_recluster_e1_draws_summary.csv`). In the balanced 1:1 design MCC isolates the matched-vs-random contrast at caliper 0.2 SD: 0.277 vs. 0.448 for random 1:1 subsets (`psm_sensitivity_caliper_sweep.csv`).

TABLE S22
DINUCLEOTIDE-SHUFFLED NEGATIVES: THE PAPER'S DRAW AND THE RANGE OVER 16 `PYTHONHASHSEED` DRAWS OF THE SHUFFLE. SOURCE: `shuffled_neg.csv`, `alt_model_shuffle_hashseed.csv`.

	TATA motif-only AUROC					
Data	real neg.	shuffled (range)	GC-content AUROC	*k*-mer AUROC (range)	*k*-mer MCC (range)	= paper
`cohn`	0.371	0.551 (0.549–0.554)	0.500	0.955 (0.951–0.955)	0.769 (0.759–0.776)	0/16
`nt_enhancers`	0.393	0.507 (0.503–0.510)	0.500	0.856 (0.839–0.879)	0.545 (0.520–0.627)	0/16
control	0.539	0.749 (0.746–0.754)	0.500	0.989 (0.988–0.996)	0.922 (0.906–0.937)	0/16

Shuffles of the positives remove the mono- and dinucleotide composition gap by construction (GC-content AUROC 0.500 on every task; the GC-content AUROC is that of per-sequence GC content, not a motif score). The negative-class TATA skew disappears (`cohn` TATA AUROC slightly above 0.5 in every draw), while the positives remain separable from their own shuffles by higher-order structure; for the control, whose TATA is a positive-class motif, the TATA AUROC rises. All sequences are scanned. The shuffle depends on `PYTHONHASHSEED`; the paper's exact draw was not recovered by the 16-seed scan.

TABLE S23
CONTROL-MOTIF PANEL (MOTIF-ONLY AUROC AT 1.3 BITS/POS; MEAN ± SD OVER THE PANEL, OR ESTIMATE [95% CI]) AND THRESHOLD INVARIANCE OF THE TATA SKEW. SOURCE: `route3_specificity.csv`, `route3_threshold_invariance.csv`.

Data	Motif	Motif-only AUROC
`cohn`	real TATA (TBP PSSM)	0.367 [0.356, 0.378]
`cohn`	column-scrambled TBP (20 PWMs)	0.376 ± 0.017
`cohn`	IC-matched random PWMs (10)	0.502 ± 0.033
`nt_enhancers`	real TATA (TBP PSSM)	0.392 [0.383, 0.400]
`nt_enhancers`	column-scrambled TBP (20 PWMs)	0.396 ± 0.027
`nt_enhancers`	IC-matched random PWMs (10)	0.507 ± 0.016
`nt_prom_tata`	real TATA (TBP PSSM)	0.589 [0.577, 0.600]
`cohn`	TATA AUROC over 0.8–1.5 bits/pos: original / equalized	0.307–0.401 / 0.485–0.503
`nt_enhancers`	TATA AUROC over 0.8–1.5 bits/pos: original / equalized	0.282–0.431 / 0.490–0.501

An AUROC below 0.5 means the motif is commoner in the negatives. Random PWMs matched to the TBP information content produce no class skew, so the scan does not manufacture one; column-scrambled TBP PWMs keep the TBP columns' AT-rich composition but not the ordered TATA pattern and stay skewed, so the skew is composition-driven. Real TATA on `nt_promoter_tata` is skewed toward the positives (positive control). The panel was re-run on homology-clean data (`homology_recluster_extra_specificity.csv`); see Table S35. Both panels scan a seeded sample of at most 4,000 sequences per class (the equalized splits keep fewer negatives); on all sequences the real-TATA AUROC of the original split is 0.371 (`cohn`) and 0.393 (`nt_enhancers`; Table S22).

S5. FM PROBES, FROZEN AND FINE-TUNED

S5.1 Frozen probes

A frozen probe's *test effect* is the AUROC change of the same fixed head when composition-biased test negatives are removed (negative = loss), leaving the GC-matched subset (test negatives subsampled to the positives' GC histogram; the paper's GC-matched intervention). It is significant for both FMs on both contaminated sets, for both heads and under both poolings (Table S24, Fig. S6). Whether the GC-balanced control is flat depends on the pooling: it is flat under HyenaDNA's [SEP] token (+0.005), not under mean pooling (–0.039). The small test effect is a floor: on composition-equalized negatives the same models lose 0.13–0.21 `cohn` AUROC, frozen or fine-tuned (Table S27), a loss that also includes genuine composition signal.

TABLE S24
FROZEN FM PROBES: TEST EFFECT, THE AUROC CHANGE OF THE SAME FIXED HEAD (NEGATIVE = LOSS) WHEN COMPOSITION-BIASED TEST NEGATIVES ARE REMOVED (GC-MATCHED ARM), BY POOLING AND HEAD. SOURCE: `fm_head_invariance.csv`, `fm_cross_architecture.csv`.

Model	Data	Pooling	Head	Test effect [95% CI]	CI excludes 0
HyenaDNA-tiny	`cohn`	[SEP]	LightGBM	−0.0481 [−0.053, −0.043]	yes
HyenaDNA-tiny	`cohn`	[SEP]	LR	−0.0502 [−0.056, −0.045]	yes
HyenaDNA-tiny	`cohn`	mean	LightGBM	−0.0575 [−0.063, −0.052]	yes
HyenaDNA-tiny	`cohn`	mean	LR	−0.0727 [−0.079, −0.067]	yes
HyenaDNA-tiny	`nt_enhancers`	[SEP]	LightGBM	−0.0793 [−0.107, −0.056]	yes
HyenaDNA-tiny	`nt_enhancers`	[SEP]	LR	−0.0971 [−0.128, −0.067]	yes
HyenaDNA-tiny	`nt_enhancers`	mean	LightGBM	−0.0799 [−0.108, −0.055]	yes
HyenaDNA-tiny	`nt_enhancers`	mean	LR	−0.1243 [−0.159, −0.093]	yes
HyenaDNA-tiny	control	[SEP]	LightGBM	+0.0055 [−0.003, +0.014]	no
HyenaDNA-tiny	control	[SEP]	LR	+0.0040 [−0.004, +0.013]	no
HyenaDNA-tiny	control	mean	LightGBM	−0.0385 [−0.047, −0.030]	yes
HyenaDNA-tiny	control	mean	LR	−0.0188 [−0.033, −0.006]	yes
DNABERT-2-117M	`cohn`	mean	LightGBM	−0.0467 [−0.051, −0.042]	yes
DNABERT-2-117M	`cohn`	mean	LR	−0.0486 [−0.053, −0.044]	yes
DNABERT-2-117M	`cohn`	[CLS]	LightGBM	−0.0465 [−0.052, −0.042]	yes
DNABERT-2-117M	`cohn`	[CLS]	LR	−0.0472 [−0.052, −0.043]	yes
DNABERT-2-117M	`nt_enhancers`	mean	LightGBM	−0.0720 [−0.100, −0.049]	yes
DNABERT-2-117M	`nt_enhancers`	mean	LR	−0.0939 [−0.126, −0.065]	yes
DNABERT-2-117M	`nt_enhancers`	[CLS]	LightGBM	−0.0771 [−0.106, −0.051]	yes
DNABERT-2-117M	`nt_enhancers`	[CLS]	LR	−0.0921 [−0.125, −0.063]	yes

Nested paired bootstrap. The effect has the same sign and significance for both heads under each pooling (head-invariant). Whether the GC-balanced control (`drosophila_enhancers_stark`) is flat depends on the pooling: it is flat under the [SEP] token only. DNABERT-2 was not run on the control; its flat reference in the paper is a placebo that removes as many `cohn` test negatives at random.

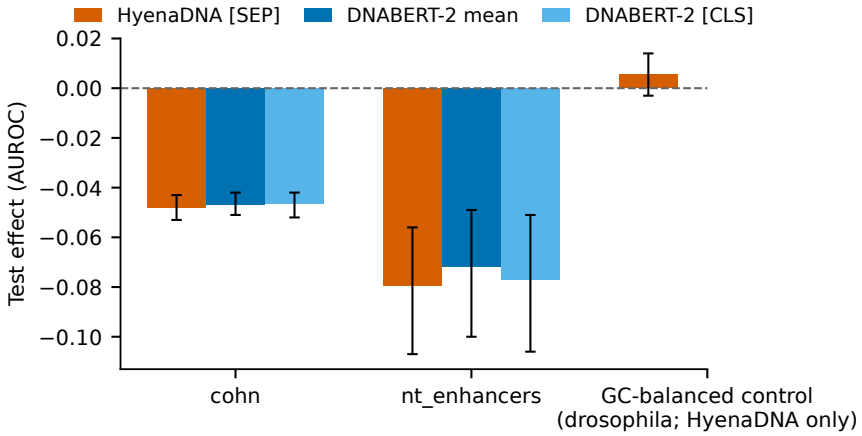


Fig. S6. Frozen FM probes: test effect (AUROC change, negative = loss; Section S5) of the fixed LightGBM head on the GC-matched arm for HyenaDNA ([SEP] token) and DNABERT-2 (mean and [CLS] pooling), with 95% bootstrap CIs (data as in Table S24). The GC-balanced control (`drosophila`) was probed with HyenaDNA only, [SEP] token. Seed 42. Re-plotted for this supplement from `fm_cross_architecture.csv`; same data as `results/upgrades/figures/fm_cross_architecture.png`.

S5.2 End-to-end fine-tuning

Two FMs were fine-tuned end to end, 54 units in total: HyenaDNA-tiny-16k-d128 [11] with all 436,354 parameters trainable (AdamW, learning rate 6×10^{-4}, 6 epochs), and DNABERT-2-117M [12] with LoRA [13] (rank 8, $\alpha = 16$, on the attention and GLU layers; 1,476,098 of 118,544,642 parameters trainable, 1.25%; AdamW, 3×10^{-4}, at most 4 epochs with early stopping). Each has a linear head on mean-pooled states and class-weighted cross-entropy; the epoch is chosen on a 10% validation carve of the training split; seeds 42–44; one fixed recipe per model, with no hyperparameter search. Every model scores the whole original test split, so the arms are the same weights scored on different test negatives: original (A), composition-equalized (B) and a size-matched random subset of the original negatives (C). Models retrained on equalized (D) or random (E) negatives are also reported. The contrasts and rules were fixed before any unit finished; the share of above-chance AUROC lost is a post hoc re-expression.

On `cohn` both models lose AUROC on equalized test negatives and nothing on random ones (Table S25): HyenaDNA

0.796 to 0.581 [0.542, 0.618] (random 0.804) and DNABERT-2 0.848 to 0.720 [0.683, 0.752] (random 0.863; by arm in Fig. S8). Re-expressed, that is 72% and 37% of the above-chance AUROC (Fig. S7). Fine-tuned HyenaDNA's share is not significantly different from the k-mer model's in the same design (66%; paired difference +0.063 [−0.007, +0.135]), and LoRA reduces but does not remove the reliance. Rule 1 is met for HyenaDNA only through its $B < 0.6$ clause and is not met for DNABERT-2 (Table S26). B is not a chance draw of a small negative set: it lies below all 1000 random draws of as many negatives (Table S30).

TABLE S25
END-TO-END FINE-TUNING (SEED MEANS OVER 42–44): THE SAME WEIGHTS SCORED ON ORIGINAL (A), COMPOSITION-EQUALIZED (B) AND SIZE-MATCHED RANDOM (C) TEST NEGATIVES; D/E: RETRAINED ON EQUALIZED/RANDOM NEGATIVES. SOURCE: `fm_finetune_v2_paper_table.csv`, `fm_finetune_v2_unit_diagnostics.csv`.

Model	Data	A	B [CI]	C	D	E	S=A−B [CI]	T=C−B [CI]	Share lost [CI] (random)
HyenaDNA-tiny, full	`cohn`	0.796	0.581 [0.542, 0.618]	0.804	0.522*	0.775	+0.214 [+0.180, +0.252]	+0.223 [+0.181, +0.271]	72% [60, 86] (−3%)
DNABERT-2, LoRA	`cohn`	0.848	0.720 [0.683, 0.752]	0.863	0.703	0.833	+0.128 [+0.095, +0.161]	+0.143 [+0.103, +0.183]	37% [27, 46] (−4%)
HyenaDNA-tiny, full	`nt_enhancers`	0.762	0.593 [0.504, 0.684]	0.797	0.643	0.822	+0.169 [+0.095, +0.241]	+0.204 [+0.099, +0.309]	64% [35, 98] (−13%)
DNABERT-2, LoRA	`nt_enhancers`	0.820	0.680 [0.592, 0.773]	0.833	0.693	0.816	+0.140 [+0.069, +0.216]	+0.153 [+0.049, +0.265]	44% [20, 70] (−4%)
HyenaDNA-tiny, full	control	0.828	0.742 [0.707, 0.779]	0.846	0.726	0.819	+0.086 [+0.056, +0.114]	+0.105 [+0.065, +0.142]	26% [17, 35] (−6%)

AUROC; CIs from 1000 resamples of the original test sequences, jointly across arms and averaged over seeds. Share lost = $(A-B)/(A-0.5)$, the share of above-chance AUROC lost (post hoc re-expression of the pre-specified S); random share = $(A-C)/(A-0.5)$, in parentheses. * all three retrained units are degenerate one-class fits (every test probability ≥0.5), so this D is not evidence that the equalized arm is unlearnable. Rule verdicts are in Table S26.

TABLE S26
PRE-SPECIFIED FINE-TUNING DECISION RULES, AS EVALUATED. SOURCE: `fm_finetune_v2_paper_table.csv`.

Model	Data	1: rides the shortcut	2: drop not a test-imbalance artifact	3: cause of the retrained collapse	5: control not destroyed
HyenaDNA-tiny, full	`cohn`	met	met	composition removal (random arm preserves signal)	–
DNABERT-2, LoRA	`cohn`	not met	met	composition removal (random arm preserves signal)	–
HyenaDNA-tiny, full	`nt_enhancers`	met	met	composition removal (random arm preserves signal)	–
DNABERT-2, LoRA	`nt_enhancers`	not met	met	composition removal (random arm preserves signal)	–
HyenaDNA-tiny, full	control	not met	met	composition removal (random arm preserves signal)	not met

Rule 1: S>0 (CI excludes 0) and either B's CI covers 0.5 or B<0.6; for HyenaDNA it is met only through B<0.6 (B's CI excludes 0.5). Rule 2: T>0 (CI excludes 0) and C's CI overlaps A's. Rule 3 reads U=E−D and R; for HyenaDNA on `cohn` it rests on degenerate retrains (Table S25). Rule 5 (GC-balanced control not destroyed): B and D stay above chance (CIs above 0.5) and the control's S lies below that of each contaminated set (its CI upper bound below the other CI's lower bound). Not met: B and D stay above chance, but the control's S, +0.086 [+0.056, +0.114], overlaps that of `nt_enhancers`, +0.169 [+0.095, +0.241]. For `nt_enhancers` the retrain-side reading is not used, because the matching test there is not supported (Table S12).

The equalizer keeps more cCRE-overlapping negatives than the random control does (`cohn` test negatives: 50% against 32%; Table S28). Without them the `cohn` drop shrinks but persists (56% and 24% of the above-chance AUROC). On `nt_enhancers` (36 equalized test negatives) both models lose AUROC (HyenaDNA 0.762→0.593; DNABERT-2 0.820→0.680), but without cCRE-overlapping negatives the loss is no longer separable from the random control (T CIs [−0.000, +0.227] and [−0.045, +0.192]; Table S29), so the fine-tuned drop is established for `cohn` only.

The GC-balanced control also loses 26% [17, 35] of its above-chance AUROC (HyenaDNA only). The pre-specified rule 5 (the control's B and D above chance and its S below that of each contaminated set, with disjoint CIs) is not met: B and D stay above chance, but its S, +0.086 [+0.056, +0.114], overlaps that of `nt_enhancers`, +0.169 [+0.095, +0.241] (Table S26). After fine-tuning, the control's GC-matched test effect is also no longer flat (−0.025; Table S32). Fine-tuning, like matching, shows reliance on composition-associated signal, not that the signal is an artifact. Post hoc, on the leak-free `nt_enhancers` test the drop persists but carries the cCRE caveat, and on homology-clean data fine-tuned HyenaDNA still loses 0.208 [0.170, 0.245] on `cohn`, while on the homology-clean `nt_enhancers` re-split its GC-matched shortcut vanishes (S_{gc} 0.003 [−0.003, +0.010]; Table S31).

Caveats. All three HyenaDNA `cohn` models retrained on equalized negatives collapsed to one-class predictors, so their D is not evidence that the arm is unlearnable. Fine-tuned HyenaDNA reaches a `cohn` accuracy of 0.716 against 0.742 published for the same architecture after a sweep [11], and an `nt_enhancers` MCC of 0.428 against 0.626 published for a larger (1.6M) HyenaDNA [11]; the models are not leaderboard-tuned. On `nt_enhancers` 11%–22%

of validation rows have an exact copy among the fitting rows, so validation AUROC overstates test AUROC (per-model means 0.884 against 0.713 for HyenaDNA, 0.953 against 0.776 for DNABERT-2); the test split is never used for selection. The only full-parameter model is the 0.44M-parameter HyenaDNA-tiny; full-parameter DNABERT-2 and NT-v2-500M runs are packaged but not run (Section S9), and DNABERT-2 was not run on the GC-balanced control (Section S1.4).

TABLE S27
ONE DESIGN, THREE MODEL FAMILIES ON `cohn`: THE SAME TEST SETS FOR THE k-MER MODEL, THE FROZEN PROBES AND THE FINE-TUNED FMS (POST HOC, DESCRIPTIVE). SOURCE: `fm_finetune_v2_compare.csv`, `fm_finetune_v2_compare_paired.csv`.

Model	A	B [CI]	C	S	Share lost [CI]
k-mer LightGBM	0.805	0.603 [0.564, 0.639]	0.820	+0.202	66% [55, 79]
HyenaDNA frozen [SEP]	0.723	0.577 [0.536, 0.614]	0.740	+0.146	66% [50, 83]
HyenaDNA frozen mean	0.790	0.621 [0.580, 0.660]	0.802	+0.169	58% [46, 72]
HyenaDNA fine-tuned (full)	0.796	0.581 [0.542, 0.618]	0.804	+0.214	72% [60, 86]
DNABERT-2 frozen mean	0.831	0.688 [0.650, 0.724]	0.841	+0.142	43% [33, 54]
DNABERT-2 frozen [CLS]	0.804	0.672 [0.632, 0.709]	0.814	+0.132	44% [32, 56]
DNABERT-2 fine-tuned (LoRA)	0.848	0.720 [0.683, 0.752]	0.863	+0.128	37% [27, 46]
HyenaDNA fine-tuned (full) − k-mer LightGBM					+0.063 [−0.007, +0.135]
HyenaDNA fine-tuned (full) − HyenaDNA frozen [SEP]					+0.069 [−0.084, +0.218]
HyenaDNA fine-tuned (full) − HyenaDNA frozen mean					+0.141 [+0.052, +0.238]
DNABERT-2 fine-tuned (LoRA) − DNABERT-2 frozen mean					−0.061 [−0.109, −0.013]
DNABERT-2 fine-tuned (LoRA) − k-mer LightGBM					−0.293 [−0.387, −0.213]

Lower block: paired differences of the share lost (same resamples). The k-mer share here is in the fine-tuning design; the overlap-weighting design of Section S4 gives a different k-mer share, so the design is always named. Non-fine-tuned baselines are single deterministic fits on the full training arm.

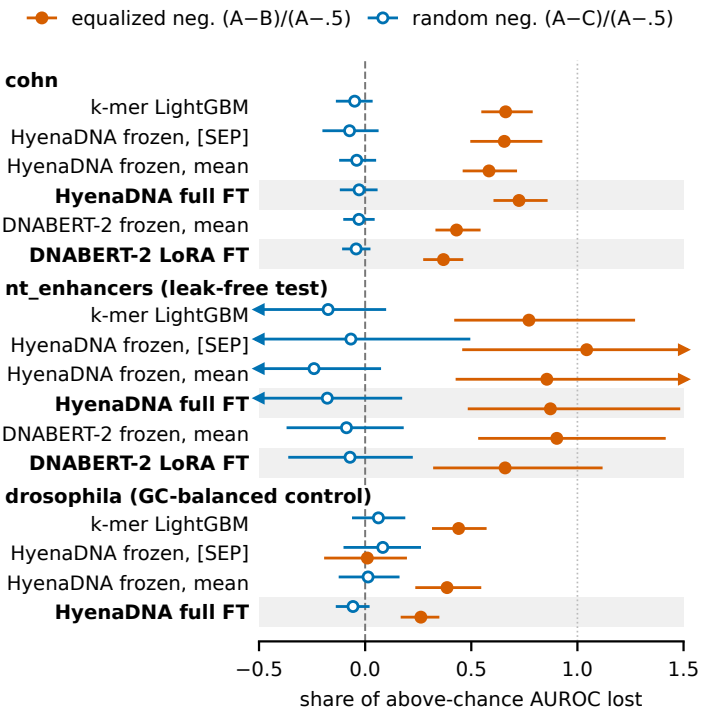


Fig. S7. Share of above-chance AUROC lost when the same weights are scored on composition-equalized test negatives, $(A-B)/(A-0.5)$ (filled), and on size-matched random test negatives, $(A-C)/(A-0.5)$ (open), for the k-mer model, frozen probes and fine-tuned FMs (post hoc, descriptive; 95% CIs). `nt_enhancers` is on its leak-free test (36 equalized negatives; wide intervals, some past 1); `drosophila` is the GC-balanced control. The cCRE caveat of Table S29 applies. Source: `results/upgrades/figures/fm_finetune_v2_compare.pdf`.

TABLE S28
EQUALIZED TEST NEGATIVES OVERLAP ENCODE CCRES MORE OFTEN THAN SIZE-MATCHED RANDOM TEST NEGATIVES (POST HOC Q3). SOURCE: `fm_finetune_v2_ccre_enrichment.csv`.

Data	Annotation	Equalized neg. overlapping	Random neg.	Positives	Fisher p
`cohn`	any cCRE	115/228 (50%)	32%	93%	6.0×10^{-5}
`cohn`	enhancer-like (ELS)	76/228 (33%)	21%	91%	0.0030
`nt_enhancers`	any cCRE	14/36 (39%)	14%	84%	0.031
`nt_enhancers`	enhancer-like (ELS)	9/36 (25%)	8%	76%	0.11

The leak-free `nt_enhancers` test has the same 36 equalized negatives. cCRE overlap is itself composition-correlated (positives overlap far more often), so removing cCRE-overlapping negatives also removes the most positive-like negatives; the cCRE-free shares below are a conservative sensitivity analysis, not the 'true' composition share.

TABLE S29
FINE-TUNED AND k-MER MODELS RE-SCORED WITHOUT CCRE-OVERLAPPING TEST NEGATIVES (POST HOC Q3). SOURCE: `fm_finetune_v2_ccre_sensitivity.csv`.

Data	Model	B neg.	B [CI]	S [CI]	T [CI]	Share lost
cohn	HyenaDNA fine-tuned (full)	113	0.654 [0.604, 0.701]	+0.194 [+0.149, +0.242]	+0.174 [+0.125, +0.228]	56% (unfiltered 72%)
	DNABERT-2 fine-tuned (LoRA)	113	0.804 [0.762, 0.842]	+0.098 [+0.063, +0.135]	+0.101 [+0.059, +0.144]	24% (unfiltered 37%)
	k-mer LightGBM	113	0.696 [0.648, 0.742]	+0.164 [+0.121, +0.210]	+0.166 [+0.116, +0.215]	46% (unfiltered 66%)
nt_enhancers	HyenaDNA fine-tuned (full)	22	0.691 [0.578, 0.789]	+0.105 [+0.020, +0.204]	+0.109 [−0.000, +0.227]	35% (unfiltered 64%)
	DNABERT-2 fine-tuned (LoRA)	22	0.774 [0.670, 0.870]	+0.078 [+0.001, +0.170]	+0.065 [−0.045, +0.192]	22% (unfiltered 44%)
	k-mer LightGBM	22	0.803 [0.726, 0.873]	+0.075 [+0.022, +0.138]	+0.095 [+0.028, +0.167]	20% (unfiltered 43%)
nt_enhancers_leakfree	HyenaDNA fine-tuned (full)	22	0.618 [0.495, 0.731]	+0.113 [+0.033, +0.211]	+0.120 [+0.008, +0.238]	49% (unfiltered 87%)
	DNABERT-2 fine-tuned (LoRA)	22	0.677 [0.559, 0.784]	+0.098 [+0.011, +0.189]	+0.091 [−0.027, +0.224]	36% (unfiltered 66%)
	k-mer LightGBM	22	0.652 [0.536, 0.762]	+0.118 [+0.040, +0.209]	+0.146 [+0.043, +0.258]	44% (unfiltered 77%)

On `cohn` both S and T exclude 0 for every model. On `nt_enhancers` (22 cCRE-free equalized negatives) S stays positive but T's CI includes 0 for both fine-tuned models, so there the loss is no longer separable from the size-matched random control; the fine-tuned drop is established for `cohn` only.

TABLE S30
IS B A CHANCE DRAW OF A SMALL NEGATIVE SET? 1000 RANDOM DRAWS OF AS MANY TEST NEGATIVES AS THE EQUALIZED ARM HOLDS, WITH THE SAME POSITIVES (POST HOC Q4). SOURCE: `fm_finetune_v2_randdraw.csv`.

Data	Model	Neg. per draw	B	Draw min (2.5–97.5%)	C	Draws ≤ B
cohn	HyenaDNA fine-tuned (full)	228	0.581	0.753 (0.768–0.824)	0.804	0
cohn	DNABERT-2 fine-tuned (LoRA)	228	0.720	0.803 (0.822–0.870)	0.863	0
nt_enhancers	HyenaDNA fine-tuned (full)	36	0.593	0.634 (0.690–0.822)	0.797	0
nt_enhancers	DNABERT-2 fine-tuned (LoRA)	36	0.680	0.702 (0.752–0.880)	0.833	0
nt_enhancers_leakfree	HyenaDNA fine-tuned (full)	36	0.525	0.582 (0.624–0.761)	0.732	0
nt_enhancers_leakfree	DNABERT-2 fine-tuned (LoRA)	36	0.582	0.620 (0.668–0.805)	0.756	0
drosophila	HyenaDNA fine-tuned (full)	217	0.742	0.784 (0.801–0.854)	0.846	0

The random draws do not control for cCRE enrichment (Table S28); for `nt_enhancers` read them together with Table S29. The GC-balanced control's B also lies below every draw.

TABLE S31
FINE-TUNING ON LEAK-FREE AND ON HOMOLOGY-CLEAN DATA (POST HOC; NO DECISION RULE ATTACHED). SOURCE: `fm_finetune_v2_posthoc_paper_table.csv`, `homology_recluster_ft_contrasts.csv`, `homology_recluster_ft_split_manifest.csv`.

Model	Data	A	B [CI]	C	S [CI]
HyenaDNA fine-tuned (full)	leak-free `nt_enhancers` (100 pos./36 neg.)	0.697	0.525 [0.424, 0.620]	0.732	+0.172 [+0.105, +0.247]
DNABERT-2 fine-tuned (LoRA)	leak-free `nt_enhancers` (100 pos./36 neg.)	0.739	0.582 [0.474, 0.682]	0.756	+0.158 [+0.079, +0.242]
HyenaDNA fine-tuned (full)	`cohn` on homology-clean data (id80; 176 eq. test neg.)	0.791	0.583 [0.541, 0.625]	0.797	+0.208 [+0.170, +0.245]
HyenaDNA fine-tuned (full)	GC-balanced control on homology-clean data (id80; 429 eq. test neg.)	0.829	0.769 [0.740, 0.796]	0.831	+0.059 [+0.043, +0.075]
HyenaDNA fine-tuned (full)	`nt_enhancers` on homology-clean data (3 eq. test neg.)	0.984	n/e	n/e	S_{gc} +0.003 [−0.003, +0.010]

Leak-free `nt_enhancers`: the 100 test positives that occur verbatim in training are removed; 14 of the 36 equalized negatives overlap a cCRE against 5 of 36 random ones (Table S28), so these rows carry that caveat. Homology-clean data: the cluster-aware id80 re-split, HyenaDNA only (DNABERT-2 on these splits is packaged, not run; Section S9). n/e: not evaluable (fewer than 10 test negatives in the arm); S_{gc} uses the GC-matched arm instead. On this re-split the fine-tuned GC-matched shortcut on `nt_enhancers` vanishes; 96.5% of its test set comes from the provided training split.

TABLE S32
GC-MATCHED TEST EFFECT $G-A$ (AUROC OF THE SAME WEIGHTS ON THE GC-MATCHED TEST SUBSET MINUS ON THE ORIGINAL TEST; POST HOC Q2).
SOURCE: `fm_finetune_v2_gc_paired.csv`.

Data	Model	$G-A$ [CI]
cohn	HyenaDNA frozen [SEP]	−0.048 [−0.053, −0.043]
cohn	HyenaDNA frozen mean	−0.058 [−0.062, −0.053]
cohn	HyenaDNA fine-tuned (full)	−0.060 [−0.065, −0.055]
cohn	DNABERT-2 frozen mean	−0.047 [−0.052, −0.042]
cohn	DNABERT-2 fine-tuned (LoRA)	−0.041 [−0.045, −0.036]
cohn	*k*-mer LightGBM	−0.059 [−0.064, −0.054]
nt_enhancers	HyenaDNA frozen [SEP]	−0.079 [−0.105, −0.053]
nt_enhancers	HyenaDNA frozen mean	−0.080 [−0.106, −0.055]
nt_enhancers	HyenaDNA fine-tuned (full)	−0.109 [−0.143, −0.080]
nt_enhancers	DNABERT-2 frozen mean	−0.072 [−0.096, −0.049]
nt_enhancers	DNABERT-2 fine-tuned (LoRA)	−0.082 [−0.110, −0.056]
nt_enhancers	*k*-mer LightGBM	−0.076 [−0.102, −0.054]
drosophila	HyenaDNA frozen [SEP]	+0.005 [−0.003, +0.014]
drosophila	HyenaDNA frozen mean	−0.039 [−0.047, −0.031]
drosophila	HyenaDNA fine-tuned (full)	−0.025 [−0.030, −0.019]
drosophila	*k*-mer LightGBM	−0.035 [−0.042, −0.028]

The frozen rows reproduce the point estimates of the frozen probes' test effect (Table S24); their CIs come from this analysis's paired bootstrap and differ slightly. After fine-tuning the GC-balanced control is no longer flat.

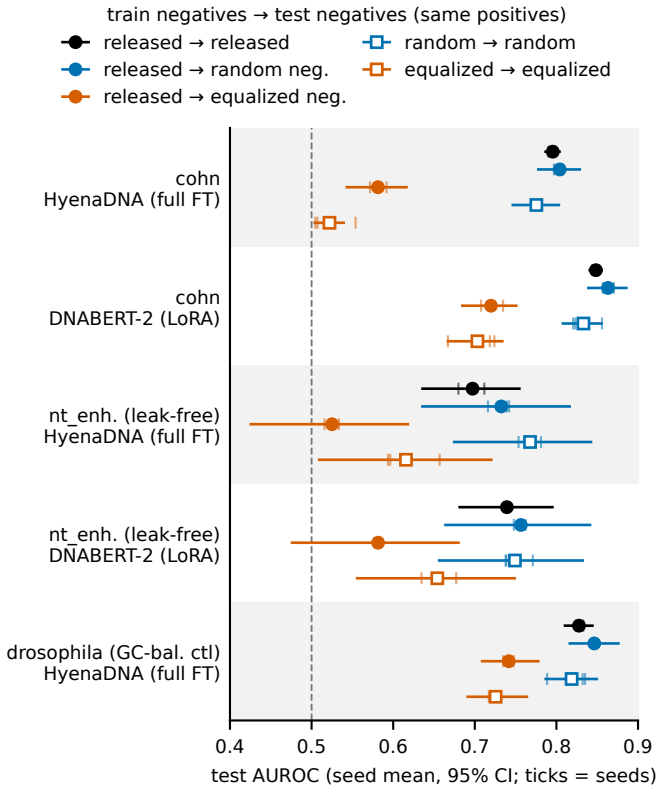


Fig. S8. Fine-tuned models: test AUROC by training arm → test negatives (seed mean, 95% CI; ticks: individual seeds; in-figure "released" denotes the original benchmark negatives). The HyenaDNA `cohn` equalized→equalized point is a degenerate one-class fit (all three seeds) and is not evidence that the arm is unlearnable. `nt_enh.` is on its leak-free test. Source: `results/upgrades/figures/fm_finetune_v2.pdf`.

S6. Homology re-clustering

All 15 tasks were re-clustered with MMseqs2 [14]: an all-vs-all nucleotide search on both strands, edges at ≥80% identity (id80, primary; id50, strict) over ≥80% of both sequences, and connected components as clusters, so near-duplicates chained through intermediates cannot straddle a split. Two protocols re-score every baseline, each against a control: a purge that scores the same trained model on the test split minus every test sequence with a training homolog, against 1000 random removals with the same per-class counts; and a cluster-aware re-split, against a random re-split, over three split seeds.

A median 11.2% of test sequences have an id80 training homolog (maximum 44.9%; `cohn` 4.3%). Against the random-removal null, the pre-specified test flags inflation in 9 of the 14 evaluable tasks; the effect exceeds 0.05 MCC (a post hoc cut) in 4, one of which (`nt_enhancers`) coincides with removing all 100 verbatim strong enhancers (Table S33; Fig. S9).

Labels (Table S34). The purge preserves every label it can score on the prevalence-adjusted MCC (the purged test set re-weighted so that each class has its original test proportion; post hoc), because purging can change the class mix, which by itself moves MCC; `human_nontata_promoters` passes only after that adjustment (raw purged MCC 0.469, adjusted 0.768). Under cluster-aware re-splits 12/15 labels are unchanged at seed 42 and 12/15 by split-seed mean. `human_nontata_promoters` relabels because of homology (MCC 0.64 against 0.87 on random re-splits), and the two NT enhancer tasks become solved under random re-splits too, a re-pooling effect of their separately built test split. The label counts are partly true by construction (composition labels are task types; once a task fails its contamination vs. shared-motif label is set by the model-free gate; the purge re-scores only the MCC and that gate), so the informative labels are the solved and positional ones.

Routes. Route 1 under homology control is in Section S2. Route 2 persists: the splice rescue is +0.62 on homology-clean data against +0.60 on the same training arm, graded in bin resolution (Table S35). For Route 3, TATA-flag cleaning of homology-clean `cohn` moves the TATA AUROC to 0.435 against 0.373 for a size-matched placebo, composition equalization stays below its size-matched placebo (k-mer AUROC 0.625 against 0.737 in one draw; 0.551–0.633 over 20 equally valid matching draws, median 0.606), the control-motif panel and the TSS depletion are preserved, and the `cohn` FM shortcut persists (frozen probes; fine-tuned HyenaDNA). On homology-clean `nt_enhancers` the frozen probes shrink but every CI still excludes 0, while the fine-tuned GC-matched shortcut vanishes (Table S31).

Limits. MMseqs2 cannot seed very short or mostly-N sequences: 316 short exact twins in `human_enhancers_ensembl` (median 15.5 bp) and 644 masked all-N sequences in `demo_human_or_worm` are reported as a sensitivity analysis, not merged into the graph. Many tests are uncorrected for multiplicity; the pre-specified rules and the named primary Route-1 test are the calls. The semi-synthetic dose-response is not re-run, by design (its random backgrounds have no natural homology), and DNABERT-2 fine-tuning on the homology-clean splits was not run (Section S9).

TABLE S33
Homology re-clustering (MMseqs2, id80): train/test homology, purge vs. random-removal null, and cluster-aware vs. random re-splits for all 15 tasks. Source: `homology_recluster_table.csv`, `homology_recluster_paper_table.csv`.

	Test sequences		Purge vs. random removal				Re-split (seeds 42–44)		
Task	homolog	exact dup.	MCC	MCC raw/adj.	effect (p)	flag	cluster/random	cl.−rand. [range]	robust
`human_or_worm`	11.2%	2.6%	0.914	0.908 / 0.908	−0.006 (<0.001)	yes	0.909 / 0.912	−0.003 [−0.009, +0.001]	no
`regulatory`	6.1%	0.0%	0.907	0.905 / 0.906	−0.001 (0.030)	no	0.911 / 0.911	0.000 [−0.000, +0.001]	no
`nt_prom_all`	3.9%	0.4%	0.888	0.888 / 0.887	−0.001 (0.33)	no	0.888 / 0.883	+0.006 [+0.002, +0.010]	no
`nt_prom_no_tata`	3.7%	0.3%	0.887	0.885 / 0.885	−0.002 (0.090)	no	0.881 / 0.886	−0.005 [−0.008, +0.000]	no
`nontata`	44.9%	0.0%	0.874	0.469 / 0.768	−0.058 (<0.001)	yes	0.638 / 0.869	−0.231 [−0.243, −0.211]	yes
`nt_prom_tata`	4.7%	1.0%	0.846	0.842 / 0.840	−0.006 (0.20)	no	0.883 / 0.869	+0.013 [−0.008, +0.040]	no
`coding`	18.8%	0.1%	0.800	0.777 / 0.777	−0.023 (<0.001)	yes	0.706 / 0.801	−0.095 [−0.137, −0.023]	yes
`enh_ensembl`	42.9%	38.0%	0.593	0.427 / 0.490	−0.089 (<0.001)	yes	0.544 / 0.596	−0.051 [−0.074, −0.037]	yes
`mouse_enh`	0.0%	0.0%	0.529	0.529 / 0.529	0.000 (1.0)	no	0.518 / 0.542	−0.024 [−0.071, +0.066]	no
`nt_enhancers`	27.0%	25.0%	0.514	0.303 / 0.317	−0.179 (<0.001)	yes	0.821 / 0.867	−0.045 [−0.063, −0.027]	no
`nt_enh_types`	27.0%	25.0%	0.476	n/e	n/e	n/e	0.769 / 0.759	+0.010 [−0.090, +0.084]	no
`cohn`	4.3%	0.0%	0.461	0.449 / 0.450	−0.012 (<0.001)	yes	0.451 / 0.460	−0.009 [−0.027, +0.002]	no
`ocr`	11.2%	0.0%	0.421	0.400 / 0.400	−0.021 (<0.001)	yes	0.400 / 0.424	−0.024 [−0.042, −0.010]	yes
control	2.1%	0.0%	0.398	0.388 / 0.388	−0.010 (<0.001)	yes	0.366 / 0.365	+0.001 [−0.014, +0.021]	no
`splice`	12.3%	7.4%	0.244	0.184 / 0.180	−0.063 (<0.001)	yes	0.195 / 0.271	−0.076 [−0.095, −0.061]	yes

Test w/ homolog: test sequences whose connected component contains a training sequence (both strands; ≥80% identity over ≥80% of both sequences). Purge: the same trained model scored on the test split minus those sequences; raw and prevalence-adjusted MCC (the purged test set re-weighted so that each class has its original test proportion; post hoc, chosen after the first results). Effect vs. null: purged minus the mean of 1000 random removals with the same per-class counts; Flag: the pre-specified homology-inflation call. For `nt_enhancers` the purge removes all 100 verbatim strong enhancers; against a subtype-matched null the effect is −0.008 (p = 0.28). Re-split: mean over split seeds 42–44; seed-robust = the pre-specified seed-42 call (CI below 0) and cluster < random in all three seeds. n/e: a class is eliminated by the purge. The 0.05-MCC materiality cut used in the text is post hoc.

TABLE S34
ROUTE LABELS AFTER HOMOLOGY CONTROL, AGAINST THE PAPER'S TABLE I LABELS. SOURCE: homology_recluster_labels_addendum.csv, homology_recluster_paper_table.csv.

Task	Table I	Purge (prev.-adj. MCC)	Cluster re-split, seed 42	Cluster re-split, seed mean	Random re-split	MCC cl./rand.	Reading
human_or_worm	composition	task type (unchanged by construction)	task type (unchanged by construction)	task type (unchanged by construction)	–	0.91/0.91	composition signal preserved
regulatory	solved	solved	solved	solved	solved	0.91/0.91	
nt_prom_all	solved	solved	solved	solved	solved	0.89/0.88	
nt_prom_no_tata	solved	solved	solved	solved	solved	0.88/0.89	
nontata	solved	solved	shared-motif	shared-motif	solved	0.64/0.87	relabel attributable to homology
nt_prom_tata	solved	solved	solved	solved	solved	0.88/0.87	
coding	composition	task type (unchanged by construction)	task type (unchanged by construction)	task type (unchanged by construction)	–	0.71/0.80	composition signal preserved
enh_ensembl	contamination (weak)	gate fires (preserved)	gate fires (CLI: Route 3)	gate fires (CLI: Route 3)	–	0.54/0.60	rests on the TATA gate alone
mouse_enh	composition	task type (unchanged by construction)	task type (unchanged by construction)	task type (unchanged by construction)	–	0.52/0.54	composition signal preserved
nt_enhancers	contamination	contamination	solved	solved	solved	0.82/0.87	also under random re-splits (re-pooling)
nt_enh_types	shared-motif	n/e	solved	solved	solved	0.77/0.76	also under random re-splits (re-pooling)
cohn	contamination	contamination	contamination	contamination	contamination	0.45/0.46	
ocr	shared-motif	shared-motif	shared-motif	shared-motif	shared-motif	0.40/0.42	
control	shared-motif	shared-motif	shared-motif	shared-motif	shared-motif	0.37/0.37	
splice	positional	positional	positional	positional	positional	0.20/0.27	

Unchanged against Table I: 12/15 at seed 42 and 12/15 by split-seed mean; the purge preserves every label it can score on the prevalence-adjusted MCC (human_nontata_promoters would relabel on the raw purged MCC). These counts are partly true by construction: the composition labels are task types; once a task fails, contamination vs. shared-motif is set by the model-free TATA gate; and the purge re-scores only the MCC and that gate. The informative labels are the solved and positional ones. A relabel is attributed to homology only if the random re-split does not reproduce it.

TABLE S35
ROUTES 2 AND 3 AND THE FROZEN FM PROBE ON HOMOLOGY-CLEAN DATA (CLUSTER-AWARE RE-SPLIT PLUS ONE REPRESENTATIVE PER COMPONENT AND LABEL). SOURCE: homology_recluster_positional.csv, homology_recluster_extra_graded.csv, homology_recluster_interventions.csv, homology_recluster_extra_cleaning.csv, homology_recluster_e1_draws_summary.csv, homology_recluster_extra_coords.csv, homology_recluster_extra_specificity.csv, homology_recluster_extra_fm.csv.

Route	Readout	Value
Route 2	splice ΔMCC (B=8−B=1), original (20,001-sequence training arm)	+0.601 [+0.571, +0.629]
Route 2	splice ΔMCC (B=8−B=1), homology-clean id80	+0.620 [+0.590, +0.651]
Route 2	splice ΔMCC (B=8−B=1), homology-clean id50	+0.653 [+0.621, +0.684]
Route 2	splice, graded over $B \in \{2, 4, 8, 16\}$ (id80)	+0.476, +0.549, +0.620, +0.672
Route 2	selectivity (pre-specified: CI includes 0 or $\lvert\Delta\rvert \leq 0.05$) on the non-positional tasks (id80)	12/13 pass
Route 2	selectivity violation: nontata (id80; a loss, not a rescue)	−0.092 [−0.125, −0.056]
Route 3	cohn TATA motif-only / GC-content AUROC, original → equalized (id80)	0.370→0.500; 0.738→0.506
Route 3	cohn TATA-flag cleaning vs. size-matched placebo (id80): TATA AUROC; k-mer MCC	0.435 vs. 0.373 (uncleaned 0.370); 0.395 vs. 0.428
Route 3	nt_enhancers TATA-flag cleaning vs. size-matched placebo (id80): TATA AUROC; k-mer MCC	0.486 vs. 0.389 (uncleaned 0.390); 0.833 vs. 0.882
Route 3	cohn equalization vs. size-matched placebo (id80; one draw, X1): k-mer AUROC	0.625 [0.582, 0.666] vs. 0.737 [0.700, 0.772]; 20 draws 0.551–0.633 (median 0.606)
Route 3	cohn TATA-flagged negatives at a TSS vs. unflagged (id80)	1.9% vs. 3.5% (×0.55; Fisher $p = 3.1\times10^{-8}$)
Route 3	control-motif panel, cohn (id80)	real TATA 0.377; scrambled TBP panel 0.377; random IC panel 0.504
Route 3	control-motif panel, nt_enhancers (id80)	real TATA 0.388; scrambled TBP panel 0.392; random IC panel 0.508
FM probe	frozen HyenaDNA [SEP], cohn: test effect (id80)	−0.051 [−0.056, −0.046]
FM probe	frozen HyenaDNA [SEP], nt_enhancers: test effect (id80)	−0.063 [−0.100, −0.030]
FM probe	frozen DNABERT-2 mean, cohn: test effect (id80)	−0.043 [−0.048, −0.039]
FM probe	frozen DNABERT-2 mean, nt_enhancers: test effect (id80)	−0.019 [−0.033, −0.008]

Equalized-arm TATA and GC-content AUROCs are partly true by construction, so the cleaning-vs.-placebo rows are the less circular replication; their k-mer MCC compares cleaning with removing as many negatives at random. The equalized k-mer AUROC depends on the matching draw: its row shows one draw (X1; equalized TATA AUROC 0.492; the TATA/GC-content row uses another, E1) and the range over 20 equally valid draws, all below the placebo's CI. The homology-clean nt_enhancers equalized arm keeps 3 test negatives at id80 (1 at id50; MCC not evaluable). GC-content AUROC: the AUROC of per-sequence GC content.

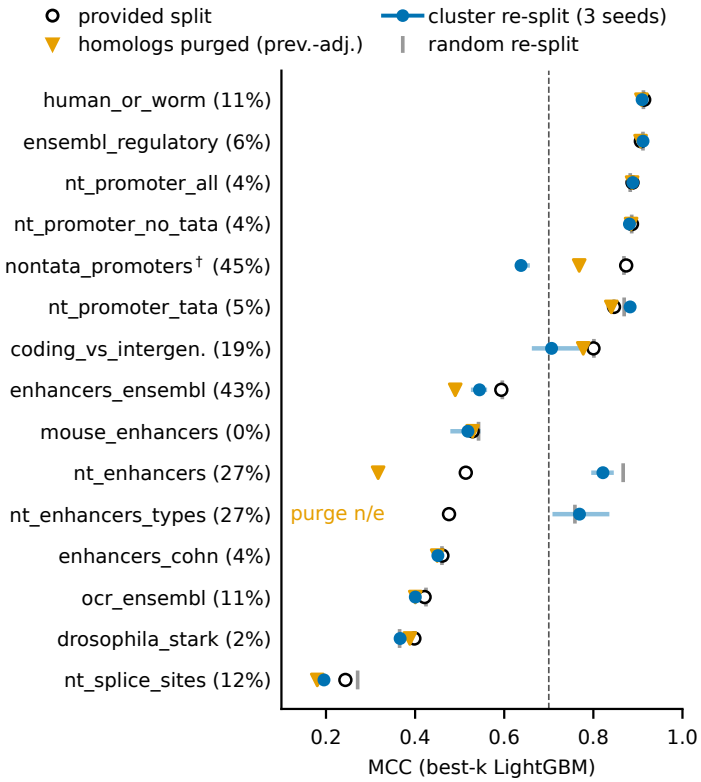


Fig. S9. MCC of the best-k LightGBM per task under homology control: provided split (open circle), test purged of training homologs (triangle; prevalence-adjusted MCC; raw purged MCC of `human_nontata_promoters` 0.469), cluster-aware re-split (filled; mean and range over split seeds 42–44) and random re-split (bar). Percent: test sequences with an id80 training homolog. Dashed line: the 0.7 "solved" threshold. The `nt_enhancers` purge removes all 100 verbatim strong enhancers; the NT-enhancer tasks rise under random re-splits as well (re-pooling). Purge n/e: a class is eliminated. Source: `results/upgrades/figures/homology_recluster_mcc.pdf`.

S7. Alternative interpretable classifiers

k-mer spectra map onto transcription-factor motifs and are the established interpretable representation for regulatory sequence [15], [16]; gradient-boosted trees capture non-linear k-mer effects on a CPU and expose the per-feature split gains that the paper's motif enrichment uses. To test whether the diagnosis depends on this classifier, every model-dependent diagnostic was refit with MCC-tuned L1/L2 logistic regression (LR; penalty and C chosen by validation MCC) and an explainable boosting machine (EBM) [17] on the 20-dimensional composition signature plus the 128 k-mers with the largest training-only ANOVA F. Only the classifier varies; the representation is the same k-mer spectrum. Model-free diagnostics (motif-only AUROCs, the TATA gate, the GC-content AUROC, coordinates, the control-motif panel) do not depend on the classifier and were not recomputed.

Against the paper's Table I labels, LR and the EBM each keep 14 of 15 labels and 7 of 7 failure-route labels (Tables S36 and S37). The counts are partly true by construction; the informative checks are the solved and positional labels, and each change is a solved task that the weaker model fails (regulatory under LR; `human_nontata_promoters` under the 128-k-mer EBM, solved again with 512 k-mers). Splice is the only positional task under every model, and the dose-response slope persists, but the real-task correlation is not significant under LR or the EBM (Table S38).

No model-level Route-3 readout is contamination-specific (Tables S39 and S40). Every task tested loses some AUROC beyond size-matched random negatives, the control included. Under overlap weighting `cohn` loses a larger share than the control under every model, but for LightGBM the composition-only baseline shows the same contrast, and under the EBM, which sees composition explicitly, the solved gate-positive `human_nontata_promoters` also loses more than the control. The weak-excess gate-positive `human_enhancers_ensembl` also loses AUROC beyond random negatives, but under overlap weighting a smaller share than the control with LightGBM and LR (with the EBM the CI of the difference includes 0); its label rests on the TATA gate alone (Table S34). Retrained against random negatives, the `cohn`-minus-control difference is significant only for the MCC-tuned LR (+0.305 [+0.091, +0.529]; AUROC-tuned LR +0.171 [−0.038, +0.390]; LightGBM +0.187 [−0.014, +0.390]; EBM +0.074 [−0.128, +0.289]). The overlap shares of

nt_enhancers exceed 1 and are not interpretable. Motif attributions agree only partly across classifiers (Table S41). Only the classifier was varied: no gapped-k-mer SVM or other representation was run in this analysis.

TABLE S36
ROUTE LABELS UNDER ALTERNATIVE INTERPRETABLE CLASSIFIERS, AGAINST THE PAPER'S TABLE I LABELS (POST HOC MAPPING). SOURCE: alt_model_agreement.csv.

Classifier	Same label	Failure-route labels kept	Changes (task: new label)
LightGBM	15/15	7/7	none
LR (MCC-tuned L1/L2)	14/15	7/7	human_ensembl_regulatory:shared-motif
EBM (128 k-mers)	14/15	7/7	human_nontata_promoters:contamination (weak)
EBM (512 k-mers)	15/15	7/7	none

Only the classifier is varied; the representation stays the k-mer spectrum at LightGBM's validation-selected k (the EBM sees the 20-d composition signature plus a train-selected k-mer subset). The counts are partly true by construction: composition labels are task types, and once a task fails its contamination vs. shared-motif label is set by the model-free TATA gate, so a weaker model can hardly change a failure label. The informative checks are the solved and positional labels: each change is a solved task that the weaker model fails.

TABLE S37
PER-TASK LABELS (MCC) UNDER EACH CLASSIFIER, MAPPED TO THE PAPER'S TABLE I VOCABULARY. SOURCE: alt_model_labels.csv.

Task	Table I	LightGBM	LR	EBM-128	EBM-512
human_or_worm	composition	composition (0.911)	composition (0.904)	composition (0.902)	composition (0.914)
regulatory	solved	solved (0.899)	**shared-motif** (0.389)	solved (0.879)	solved (0.879)
nt_prom_all	solved	solved (0.886)	solved (0.821)	solved (0.881)	solved (0.887)
nt_prom_no_tata	solved	solved (0.887)	solved (0.814)	solved (0.876)	solved (0.889)
nontata	solved	solved (0.874)	solved (0.782)	**contam. (weak)** (0.673)	solved (0.717)
nt_prom_tata	solved	solved (0.846)	solved (0.836)	solved (0.894)	solved (0.898)
coding	composition	composition (0.800)	composition (0.789)	composition (0.737)	composition (0.785)
enh_ensembl	contam. (weak)	contam. (weak) (0.569)	contam. (weak) (0.570)	contam. (weak) (0.481)	contam. (weak) (0.498)
mouse_enh	composition	composition (0.529)	composition (0.488)	composition (0.554)	composition (0.546)
nt_enhancers	contamination	contamination (0.514)	contamination (0.421)	contamination (0.449)	contamination (0.473)
nt_enh_types	shared-motif	shared-motif (0.476)	shared-motif (0.347)	shared-motif (0.348)	shared-motif (0.428)
cohn	contamination	contamination (0.461)	contamination (0.453)	contamination (0.423)	contamination (0.443)
ocr	shared-motif	shared-motif (0.399)	shared-motif (0.386)	shared-motif (0.349)	shared-motif (0.366)
control	shared-motif	shared-motif (0.398)	shared-motif (0.376)	shared-motif (0.372)	shared-motif (0.385)
splice	positional	positional (0.243)	positional (0.203)	positional (0.229)	positional (0.249)

Bold: a label that differs from Table I. MCC on the pre-specified capped training arm (tasks capped at 50k training sequences); uncapped refits change no label.

TABLE S38
ROUTES 1 AND 2 UNDER EACH CLASSIFIER. SOURCE: alt_model_route2.csv, alt_model_dose.csv, alt_model_route1.csv.

Classifier	Splice ΔMCC (B=8−1)	Largest other $\lvert\Delta\rvert$	Dose slope (pooled)	Real-task ρ, n=11	Real-task ρ, n=9†	Presence ρ (GB, n=9)
LightGBM	+0.601 [+0.571, +0.629]	0.028	+1.99 [+1.92, +2.06]	+0.74 (p 0.013)	+0.87 (p 0.0045)	−0.03
LR (MCC-tuned L1/L2)	+0.681 [+0.654, +0.711]	0.035	+1.95 [+1.88, +2.01]	+0.27 (p 0.42)	+0.13 (p 0.74)	−0.15
EBM (128 k-mers)	+0.440 [+0.409, +0.470]	0.031	+1.81 [+1.73, +1.87]	+0.47 (p 0.14)	+0.47 (p 0.21)	−0.27

Splice rescue on the same 20,001-sequence training arm for every model; the EBM with position-binned k-mers guaranteed (post hoc) gives +0.450. Largest other $\lvert\Delta\rvert$: largest CI-significant change on the other tasks. Dose slope: semi-synthetic dose-response, 135 settings. Real-task ρ: uncapped MCCs where available (the EBM keeps its capped MCC on the regulatory task). † Without human_nontata_promoters and nt_enhancers. Presence: Spearman of MCC with the number of enriched TFs.

TABLE S39
ROUTE-3 MODEL-LEVEL READOUTS UNDER EACH CLASSIFIER (POST HOC). SOURCE: `alt_model_route3_sizecontrolled.csv`, `alt_model_route3_overlap_contrast.csv`, `psm_sensitivity_between_dataset.csv`.

Task	LightGBM	LR (MCC-tuned L1/L2)	EBM (128 *k*-mers)
AUROC loss beyond size-matched random negatives (random − matched)			
`cohn`	+0.152 [+0.110, +0.193]	+0.170 [+0.124, +0.212]	+0.132 [+0.091, +0.172]
`nt_enhancers` (leak-free test)	+0.087 [−0.035, +0.217]	+0.062 [−0.045, +0.167]	+0.093 [−0.034, +0.225]
control	+0.101 [+0.063, +0.142]	+0.090 [+0.047, +0.133]	+0.108 [+0.065, +0.149]
`enh_ensembl`	+0.026 [+0.018, +0.034]	+0.050 [+0.041, +0.059]	+0.037 [+0.027, +0.047]
`ocr`	+0.036 [+0.027, +0.046]	+0.045 [+0.036, +0.054]	+0.045 [+0.035, +0.054]
Overlap weighting, model fixed: share of above-chance AUROC lost, task − GC-balanced control			
`cohn`	+0.353 [+0.293, +0.413]	+0.292 [+0.228, +0.353]	+0.419 [+0.350, +0.487]
`nt_enhancers`	not interpretable	not interpretable	not interpretable
`enh_ensembl`	−0.122 [−0.169, −0.079]	−0.137 [−0.195, −0.087]	−0.047 [−0.103, +0.004]
`ocr`	−0.112 [−0.158, −0.068]	−0.138 [−0.195, −0.088]	−0.074 [−0.132, −0.022]
`nontata`	−0.229 [−0.276, −0.187]	−0.090 [−0.150, −0.037]	+0.201 [+0.134, +0.260]

Loss beyond random: random-negative AUROC minus AUROC on the released equalized split. Every task tested loses some AUROC beyond random negatives, the GC-balanced control included. Overlap contrast: share of above-chance AUROC the unretrained model loses under overlap weighting, task minus control; for `nt_enhancers` the weights over-correct (weighted AUROC below 0.5, share above 1), so its share is not interpretable. The LightGBM contrast for `cohn` is matched by the composition-only model's contrast under the same weights (+0.352; Table S17), so these contrasts measure reliance on the composition gap, not contamination specificity. Under the EBM, which sees composition explicitly, the solved gate-positive `human_nontata_promoters` also loses more than the control.

TABLE S40
RETRAINED RANDOM-NEGATIVE DESIGN: SHARE OF ABOVE-CHANCE AUROC LOST, `cohn` MINUS THE GC-BALANCED CONTROL (POST HOC). SOURCE: `alt_model_route3_contrast.csv`.

Classifier	`cohn`	Control	Difference [CI]
LightGBM	0.573	0.386	+0.187 [−0.014, +0.390]
LR (MCC-tuned L1/L2)	0.662	0.358	+0.305 [+0.091, +0.529]
LR (AUROC-tuned)	0.563	0.392	+0.171 [−0.038, +0.390]
EBM (128 *k*-mers)	0.500	0.427	+0.074 [−0.128, +0.289]

The difference excludes 0 only for the MCC-tuned LR; with the AUROC-tuned LR it does not. No model-level readout here is contamination-specific.

TABLE S41
MOTIF ATTRIBUTION UNDER EACH CLASSIFIER (POST HOC): THE PAPER'S NAMED TFS ON `human_nontata_promoters`, AND AGREEMENT WITH LIGHTGBM ACROSS TASKS. SOURCE: `alt_model_attribution.csv`.

Classifier	SP1 rank	SP2 rank	Jaccard vs. LightGBM (nontata)	Median Jaccard (all tasks)	Median z-profile ρ
LightGBM	1	2	–	–	–
LR (MCC-tuned L1/L2)	6	4	0.49	0.25	0.76
EBM (128 *k*-mers)	4	8	0.55	0.37	0.83

Ranks among FDR-enriched vertebrate JASPAR TFs by importance-weighted enrichment; Jaccard: overlap of enriched TF sets.

S8. THE GUE SUITE

The unchanged pipeline and route thresholds were applied to the 28 datasets of GUE [12] (human, mouse, yeast and virus), with predictions stated before the run (Table S44). GUE is separately assembled but mostly not new DNA (Table S42): only 6 of the 28 datasets are near-disjoint from GB/NT (at most 0.2% of sequences shared; only covid shares none), 20–45% of the human-TF sequences occur in GB/NT, and the promoter positives, most splice sequences and all yeast sequences are NT data.

The rule gives 12 failure routes (3 positional, 5 shared-motif, 4 contamination), 8 solved and 8 "hard" labels (MCC below 0.7 with no single route; Table S43). Every failure route falls on a dataset that shares at least 65.8% of its sequences with GB/NT. The positional rule fires on splice (ΔMCC +0.407) and, against our pre-stated prediction, also on both TATA-promoter tasks (+0.231 and +0.248). `prom_300_tata` has the same positives as `nt_promoter_tata`, yet on `nt_promoter_tata` the same code gives +0.019 [−0.001, +0.042], not significant: with identical positives, the route follows how the negatives were built. (The GUE negatives of the two TATA tasks differ from NT's, although 31.6%/54.8% of their label-0 sequences also occur in the paper's tasks.)

Only the three positional labels survive our post hoc checks. The five yeast shared-motif labels depend on a length-dependent richness threshold (Table S45), which also makes 7 short tasks "hard". The four yeast contamination labels fall

on measured low-signal genomic windows (the low-signal class of each histone-mark dataset), not on sampled negatives, and genomic location accounts for only 34%–53% of their TATA excess with two location strata (Table S46); they are not evidence of negative-set contamination. Of the 6 applicable pre-stated predictions, 3 failed. A split artifact outside the three routes appears in the mouse-TF tasks, whose dinucleotide-shuffle negatives were split independently of their source positives: the composition-only baseline inverts (AUROC 0.02–0.08; Table S47). When negatives are shuffles of the positives, each positive and its shuffle belong in the same split.

On the development tasks the same driver reproduces 15 of the paper's 15 MCCs, and its rule reproduces 11 of 12 Table I routes that it can emit (in-sample consistency, not calibration). The CPU k-mer baseline sits a median of 7.2 MCC points (MCC×100) below DNABERT-2 as published, over the 27 MCC datasets; the comparison only locates the baseline. GUE was difficulty-filtered by its authors, so its route distribution is not representative of benchmarks in general.

TABLE S42
GUE TASK FAMILIES: HOW THE NEGATIVES WERE BUILT, HOW MUCH OF THE DNA IS SHARED WITH GB/NT, AND THE LABELS. SOURCE: `gue_family_summary.csv`.

Family	n	Negatives / label-0 class (as documented)	Shared	Labels	k-mer	DNABERT-2
Core promoter (70 bp)	3	as prom_300, center -34..+35 bp around the TSS	50.1–77.4%	hard:2, positional:1	58.2	70.5
Promoter (300 bp)	3	non-TATA: randomly substituted promoter sequences; TATA: random non-promoter genomic sequences carrying a TATA motif	50.0–65.8%	solved:2, positional:1	66.8	84.2
TF binding, human	5	non-overlapping genomic sequences of the same length and GC content (GC-matched)	19.8–44.8%	hard:3, solved:2	63.1	70.1
Splice site (3-class)	1	Wang et al. 2019 donor/acceptor/non-site, reconstructed with adversarial negatives	68.4–68.4%	positional:1	34.7	85.0
TF binding, mouse	5	dinucleotide-shuffled positives (dinucleotide frequencies preserved)	0.0–0.2%	hard:3, solved:2	63.5	68.0
Yeast histone marks	10	Pham et al. 2005 datasets (jaist.ac.jp/~tran/nucleosome): yeast genomic windows (ORF and intergenic ChIP-chip regions) with high (label 1) vs low (label 0) relative histone occupancy/modification; no sampled negatives; GUE split 8:1:1	100.0–100.0%	shared-motif:5, contamination:4, solved:1	51.5	56.0
Covid variant (9-class)	1	9 SARS-CoV-2 variants (GISAID), 1 kb segments	0.0–0.0%	solved:1	76.8	71.0

Shared: range over the family's datasets of the fraction of sequences found in GB/NT. Labels: 'hard' = MCC below 0.7 with no single route (Table S43). k-mer and DNABERT-2: mean MCC×100 (covid: macro-F1×100) of this pipeline's k-mer LightGBM and of DNABERT-2 as published; the comparison only locates the baseline. The yeast 'negatives' are the measured low-signal class of each mark, not a sampled negative set.

TABLE S43
THE 28 GUE DATASETS UNDER THE UNCHANGED PIPELINE AND ROUTE THRESHOLDS. SOURCE: `gue_results.csv`, `gue_overlap_summary.csv`, `gue_failure_route_audit.csv`.

Dataset	Shared with GB/NT	Class	MCC [CI]	Route	Positional ΔMCC [CI]	Post hoc status
prom_core_all	52.9%	shares with the 15	0.612 [0.594, 0.632]	hard	+0.019 [+0.005, +0.033]	hard only via richness threshold
prom_core_notata	50.1%	shares with the 15	0.646 [0.626, 0.667]	hard	0.000 [−0.013, +0.014]	hard only via richness threshold
prom_core_tata	77.4%	shares with the 15	0.488 [0.412, 0.550]	positional	+0.231 [+0.170, +0.297]	unaffected
prom_300_all	51.6%	shares with the 15	0.765 [0.750, 0.781]	solved	+0.037 [+0.025, +0.049]	
prom_300_notata	50.0%	shares with the 15	0.867 [0.854, 0.880]	solved	+0.003 [−0.005, +0.010]	
prom_300_tata	65.8%	shares with the 15	0.371 [0.296, 0.439]	positional	+0.248 [+0.183, +0.317]	unaffected
human_tf_0	44.8%	shares with the 15	0.648 [0.601, 0.692]	hard	−0.004 [−0.037, +0.025]	hard only via richness threshold
human_tf_1	42.4%	shares with the 15	0.700 [0.657, 0.744]	solved	−0.018 [−0.044, +0.009]	
human_tf_2	44.4%	shares with the 15	0.567 [0.512, 0.622]	hard	+0.021 [−0.012, +0.054]	hard only via richness threshold
human_tf_3	19.8%	shares with the 15	0.479 [0.425, 0.531]	hard	−0.023 [−0.065, +0.020]	hard only via richness threshold
human_tf_4	38.3%	shares with the 15	0.759 [0.718, 0.796]	solved	+0.004 [−0.020, +0.026]	
splice_reconstructed	68.4%	shares with the 15	0.347 [0.326, 0.368]	positional	+0.407 [+0.382, +0.432]	unaffected
mouse_tf_0	0.1%	near-disjoint	0.577 [0.520, 0.633]	hard	−0.027 [−0.061, +0.010]	hard only via richness threshold
mouse_tf_1	0.0%	near-disjoint	0.768 [0.752, 0.783]	solved	−0.006 [−0.015, +0.003]	
mouse_tf_2	0.2%	near-disjoint	0.783 [0.716, 0.844]	solved	+0.023 [−0.025, +0.070]	
mouse_tf_3	0.1%	near-disjoint	0.674 [0.578, 0.766]	hard	−0.016 [−0.050, +0.012]	
mouse_tf_4	0.1%	near-disjoint	0.373 [0.331, 0.414]	hard	−0.020 [−0.051, +0.011]	hard only via richness threshold
emp_H3	100.0%	other NT tasks only	0.689 [0.651, 0.726]	contamination	+0.005 [−0.018, +0.030]	low-signal windows
emp_H3K14ac	100.0%	other NT tasks only	0.473 [0.442, 0.504]	shared-motif	−0.060 [−0.087, −0.033]	threshold-dependent
emp_H3K36me3	100.0%	other NT tasks only	0.499 [0.468, 0.527]	contamination	−0.003 [−0.027, +0.018]	low-signal windows
emp_H3K4me1	100.0%	other NT tasks only	0.414 [0.382, 0.446]	contamination	−0.025 [−0.054, +0.001]	low-signal windows
emp_H3K4me2	100.0%	other NT tasks only	0.342 [0.309, 0.375]	shared-motif	−0.032 [−0.063, −0.003]	threshold-dependent
emp_H3K4me3	100.0%	other NT tasks only	0.385 [0.354, 0.415]	shared-motif	−0.079 [−0.110, −0.050]	threshold-dependent
emp_H3K79me3	100.0%	other NT tasks only	0.627 [0.598, 0.657]	contamination	−0.017 [−0.036, +0.002]	low-signal windows
emp_H3K9ac	100.0%	other NT tasks only	0.535 [0.502, 0.567]	shared-motif	−0.043 [−0.070, −0.017]	threshold-dependent
emp_H4	100.0%	other NT tasks only	0.723 [0.687, 0.758]	solved	−0.002 [−0.024, +0.018]	
emp_H4ac	100.0%	other NT tasks only	0.463 [0.432, 0.494]	shared-motif	−0.062 [−0.090, −0.038]	threshold-dependent
covid	0.0%	near-disjoint	0.736 [0.727, 0.746]	solved	−0.022 [−0.028, −0.017]	

Shared: fraction of the dataset's sequences that occur in GB or NT tasks as exact matches (either strand; containment in either direction). Class: 'shares with the 15', shares sequences with the paper's 15 tasks; 'other NT tasks only', shares sequences only with NT tasks outside those 15; near-disjoint, at most 0.2% shared (mouse TF 2–23 sequences per dataset; only covid shares none). Route 'hard': MCC below 0.7 and none of the three route rules fires (no single route). Post hoc status of the failure routes: the yeast (EMP) shared-motif labels depend on a length-dependent richness threshold (Table S45); the yeast contamination labels fall on measured low-signal genomic windows, not sampled negatives (Table S46); only the three positional labels are unaffected. Positional deltas other than the three Route-2 tasks are not corrected for multiplicity. covid is 9-class (macro F1 is the GUE metric).

TABLE S44
PREDICTIONS STATED BEFORE THE GUE RUN, AND THEIR OUTCOMES. SOURCE: `gue_predictions.csv`.

	Prediction	Values	Outcome
P1a	mouse TF (dinucleotide-shuffled negatives): composition-only AUROC within 0.05 of 0.5	mouse_tf_0=0.022, mouse_tf_1=0.081, mouse_tf_2=0.018, mouse_tf_3=0.033, mouse_tf_4=0.034	FAILS
P1b	mouse TF: composition fraction ~0 (read as |fraction| <= 0.10)	mouse_tf_0=-1.288, mouse_tf_1=-0.914, mouse_tf_2=-1.036, mouse_tf_3=-1.134, mouse_tf_4=-1.784	FAILS
P1c	mouse TF: equalized split retains most (>50%) negatives	mouse_tf_0=0.484, mouse_tf_1=0.793, mouse_tf_2=0.529, mouse_tf_3=0.512, mouse_tf_4=0.740	PARTLY
P2	human TF (GC-matched negatives): GC-content AUROC within 0.05 of 0.5	human_tf_0=0.504, human_tf_1=0.511, human_tf_2=0.497, human_tf_3=0.498, human_tf_4=0.503	HOLDS
P3a	splice/reconstructed: significant positive positional delta	delta=+0.4069 [+0.3820,+0.4316]	HOLDS
P3b	splice is the only GUE task that can meet the Route-2 rule (significant delta > 0.2)	tasks meeting the delta rule: prom_core_tata (+0.231), prom_300_tata (+0.248), splice_reconstructed (+0.407)	FAILS
P4	no magnitude prediction for the promoter tasks	none	not applicable

Of the 6 applicable predictions (P4 is not applicable), 2 held, 1 held partly and 3 failed: Route 2 is not confined to splice (P3b), and the mouse-TF composition-only baseline is far from chance (P1a, P1b), because shuffle twins were split across train and test (Table S47).

TABLE S45
ROUTE-1 RICHNESS THRESHOLD NORMALIZED BY SEQUENCE LENGTH (POST HOC PH4): HOW MANY LABELS CHANGE. SOURCE: `gue_rule_sensitivity_summary.csv`.

Variant	Changed	hard→shared	shared→hard	Label counts / agreement
as registered	0	0	0	hard (no single route):8; solved:8; shared-motif:5; contamination:4; positional:3
richness per 200 bp	11	6	5	solved:8; hard (no single route):7; shared-motif:6; contamination:4; positional:3
richness per 250 bp	12	7	5	solved:8; shared-motif:7; hard (no single route):6; contamination:4; positional:3
richness per 300 bp (= median paper-15 length)	11	7	4	shared-motif:8; solved:8; hard (no single route):5; contamination:4; positional:3
richness per 400 bp	7	7	0	shared-motif:12; solved:8; contamination:4; positional:3; hard (no single route):1
richness per 500 bp	7	7	0	shared-motif:12; solved:8; contamination:4; positional:3; hard (no single route):1
GB+NT (paper 15): as registered	0	0	0	Table I agreement 11/12
GB+NT (paper 15): richness per 200 bp	1	0	1	Table I agreement 10/12
GB+NT (paper 15): richness per 250 bp	0	0	0	Table I agreement 11/12
GB+NT (paper 15): richness per 300 bp (= median paper-15 length)	0	0	0	Table I agreement 11/12
GB+NT (paper 15): richness per 400 bp	0	0	0	Table I agreement 11/12
GB+NT (paper 15): richness per 500 bp	0	0	0	Table I agreement 11/12

The absolute richness threshold is unreachable for the shortest GUE tasks, which is why 7 of the 8 GUE 'hard' labels are hard; normalizing it moves labels in both directions only at 200–300 bp. GB/NT lengths lie near the reference length, so their stability is partly mechanical.

TABLE S46
YEAST (EMP) DATASETS WITH THE ROUTE-3 SIGNATURE: WHERE THE LABEL-0 WINDOWS COME FROM, AND HOW MUCH OF THE TATA EXCESS GENOMIC LOCATION ACCOUNTS FOR (POST HOC PH5). SOURCE: `gue_emp_provenance.csv`.

Dataset	Label	Label 1/0	= Pham	Excess	Location-std. (2 strata) [CI]	Accounted	4 strata	Accounted
emp_H3	contamination	7,667/7,298	yes	+0.270	+0.175 [+0.147, +0.203]	35%	+0.190	30%
emp_H4	solved	6,480/8,121	yes	+0.332	+0.198 [+0.160, +0.235]	40%	+0.200	40%
emp_H3K4me1	contamination	17,266/14,411	yes	+0.133	+0.062 [+0.038, +0.088]	53%	+0.078	41%
emp_H3K36me3	contamination	18,892/15,988	yes	+0.178	+0.092 [+0.067, +0.118]	48%	+0.061	65%
emp_H3K79me3	contamination	15,337/13,500	yes	+0.262	+0.172 [+0.144, +0.198]	34%	+0.160	39%

Label counts equal the positive/negative counts of the Pham et al. 2005 release on every mark, so label 0 is the measured low-signal class (low modification relative to H3 for the modification marks), not a sampled negative set. Excess: TATA negative-class excess on the uniquely located windows; location-standardized: the same within ORF/intergenic strata (2 strata) or finer strata (4). 'Accounted' = share of the excess removed by standardization. With 2 strata the excess falls below the Route-3 threshold (0.10; CI upper bound) for 1 of 5 marks (emp_H3K4me1); with 4 strata for 0.

TABLE S47
GUE MOUSE-TF SPLIT ARTIFACT: DINUCLEOTIDE-SHUFFLE NEGATIVES WERE SPLIT INDEPENDENTLY OF THEIR SOURCE POSITIVES (POST HOC PH2). SOURCE: `gue_twins.csv`.

Dataset	Test n	Test w/ cross-label twin in train	Comp.-only AUROC, all	with twin	without twin
mouse_tf_0	810	81.9%	0.023	0.000	0.520
mouse_tf_1	6,745	80.5%	0.081	0.015	0.516
mouse_tf_2	328	81.1%	0.018	0.000	0.493
mouse_tf_3	239	72.8%	0.033	0.000	0.452
mouse_tf_4	1,883	80.6%	0.034	0.000	0.458

A test sequence whose shuffle twin (same composition, opposite label) is in the training split inverts a composition-only model; on test sequences without such a twin the composition-only AUROC is close to 0.5. When negatives are shuffles of the positives, each positive and its shuffle belong in the same split.

S9. Analyses packaged but not run (not results)

Table S48 lists analyses that were packaged or recommended but were not run. None of them produced a number in the paper or in this supplement; the paper states the corresponding gaps as limitations (full-parameter fine-tuning covers only the 0.44M-parameter HyenaDNA-tiny; the GC-balanced control covers HyenaDNA only; homology-clean fine-tuning covers HyenaDNA only).

TABLE S48
Analyses packaged or recommended but NOT run. No number in the paper or in this supplement comes from them. Source: `src/upgrades/fm_finetune/gpu_deferred/README.md`, `src/upgrades/homology_recluster/deferred/README.md`, `src/upgrades/alt_model/deferred/README.md`.

Analysis (not run)	Why not run	Package / command
DNABERT-2-117M full-parameter fine-tuning, `cohn` and `nt_enhancers` (3 arms × seeds 42–44; 18 units)	full-model optimizer state exceeds the laptop's GPU memory	`fm_finetune/gpu_deferred/jobs.sh` 2 and 7
NT-v2-500M LoRA fine-tuning, `cohn`, `nt_enhancers` and the GC-balanced control (27 units)	0.5B parameters; days on the laptop	`fm_finetune/gpu_deferred/jobs.sh` 3, 6 and 5
DNABERT-2 LoRA fine-tuning on the homology-clean splits (`cohn`, `nt_enhancers`; 18 units)	memory above the per-job budget	`homology_recluster/deferred/run_deferred_dnabert_hc.sh`
Uncapped multiclass EBM on `human_ensembl_regulatory` (1 unit)	about 9–10 h at 3 threads; cannot change a label	`alt_model/deferred/run_ebm_regulatory_full.sh`
DNABERT-2 frozen probe and LoRA fine-tuning on the GC-balanced control	not packaged (the exclusion was a choice; Table S3)	–
Seed-pinned re-draw of the dinucleotide-shuffled negatives	not packaged (single-draw dependence is disclosed instead; Table S22)	–

GPU bundle for the fine-tuning jobs: `external/fm_finetune/gbkb_ft_gpu_bundle.tar.gz`, SHA-256 `7e6707d9583c30c7`…(verified). Paths are under `src/upgrades/`. After a GPU hand-back, the torch-free scoring chain in `src/upgrades/fm_finetune/README.md` (step 3) re-scores every unit with the same pre-specified rules.

AI-use disclosure

Claude (Anthropic), run as Claude Code agents under the author's direction, wrote the analysis and table-generating code, ran the analyses, and drafted this supplement. The author designed the study, inspected and verified the outputs, and takes full responsibility for the content.

References

[1] D. Cohn, O. Zuk, and T. Kaplan, "Enhancer identification using transfer and adversarial deep learning of DNA sequences," bioRxiv 264200, 2018, doi:10.1101/264200.

[2] B. Liu, L. Fang, R. Long, X. Lan, and K.-C. Chou, "iEnhancer-2L: a two-layer predictor for identifying enhancers and their strength by pseudo k-tuple nucleotide composition," *Bioinformatics*, vol. 32, no. 3, pp. 362–369, 2016.

[3] J. Ernst *et al.*, "Mapping and analysis of chromatin state dynamics in nine human cell types," *Nature*, vol. 473, no. 7345, pp. 43–49, 2011.

[4] B. Liu, K. Li, D.-S. Huang, and K.-C. Chou, "iEnhancer-EL: identifying enhancers and their strength with ensemble learning approach," *Bioinformatics*, vol. 34, no. 22, pp. 3835–3842, 2018.

[5] Q. Geng, R. Yang, and L. Zhang, "A deep learning framework for enhancer prediction using word embedding and sequence generation," *Biophys. Chem.*, vol. 286, p. 106822, 2022.

[6] M. Costantini, O. Clay, F. Auletta, and G. Bernardi, "An isochore map of human chromosomes," *Genome Res.*, vol. 16, no. 4, pp. 536–541, 2006.

[7] P. R. Rosenbaum and D. B. Rubin, "The central role of the propensity score in observational studies for causal effects," *Biometrika*, vol. 70, no. 1, pp. 41–55, 1983.

[8] F. Li, K. L. Morgan, and A. M. Zaslavsky, "Balancing covariates via propensity score weighting," *J. Am. Stat. Assoc.*, vol. 113, no. 521, pp. 390–400, 2018.

[9] ENCODE Project Consortium *et al.*, "Expanded encyclopaedias of DNA elements in the human and mouse genomes," *Nature*, vol. 583, no. 7818, pp. 699–710, 2020.

[10] S. F. Altschul and B. W. Erickson, "Significance of nucleotide sequence alignments: a method for random sequence permutation that preserves dinucleotide and codon usage," *Mol. Biol. Evol.*, vol. 2, no. 6, pp. 526–538, 1985.

[11] E. Nguyen *et al.*, "HyenaDNA: long-range genomic sequence modeling at single nucleotide resolution," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023.

[12] Z. Zhou, Y. Ji, W. Li, P. Dutta, R. Davuluri, and H. Liu, "DNABERT-2: efficient foundation model and benchmark for multi-species genomes," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024, arXiv:2306.15006.

[13] E. J. Hu *et al.*, "LoRA: low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, arXiv:2106.09685.

[14] M. Steinegger and J. Söding, "MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets," *Nat. Biotechnol.*, vol. 35, no. 11, pp. 1026–1028, 2017.

[15] D. Lee, R. Karchin, and M. A. Beer, "Discriminative prediction of mammalian enhancers from DNA sequence," *Genome Res.*, vol. 21, no. 12, pp. 2167–2180, 2011.

[16] M. Ghandi, D. Lee, M. Mohammad-Noori, and M. A. Beer, "Enhanced regulatory sequence prediction using gapped k-mer features," *PLoS Comput. Biol.*, vol. 10, no. 7, p. e1003711, 2014.

[17] Y. Lou, R. Caruana, J. Gehrke, and G. Hooker, "Accurate intelligible models with pairwise interactions," in *Proc. ACM SIGKDD*, 2013, pp. 623–631.

Manifest of input files

TABLE S49

Every input file read by `make_sm_tables.py`, with its SHA-256 prefix and row count. Source: this generator.

File	SHA-256 (first 12)	Rows
REPRODUCIBILITY.md	3020831ff3b1	
external/fm_finetune/gbkb_ft_gpu_bundle.tar.gz	7e6707d9583c	
results/contamination_analysis.csv	22db00a037b5	20
results/nt/contamination_analysis.csv	8725a2778ab1	20
results/upgrades/alt_model_agreement.csv	e93719c627fb	48
results/upgrades/alt_model_attribution.csv	d7426dffbf1f	45
results/upgrades/alt_model_dose.csv	5bd30314af69	12
results/upgrades/alt_model_labels.csv	0b91c9ef9c4f	180
results/upgrades/alt_model_route1.csv	e0770322382f	46
results/upgrades/alt_model_route2.csv	9c532061c83d	76
results/upgrades/alt_model_route3_contrast.csv	5026ae564fdc	20
results/upgrades/alt_model_route3_overlap_contrast.csv	90c6a13628ab	32
results/upgrades/alt_model_route3_sizecontrolled.csv	b08272a45ee8	48
results/upgrades/alt_model_shuffle_hashseed.csv	334181785825	48
results/upgrades/figures/fm_finetune_v2.pdf	29846dc6927e	
results/upgrades/figures/fm_finetune_v2_compare.pdf	b74daa287d56	
results/upgrades/figures/homology_recluster_mcc.pdf	e4602aaaa6dc	
results/upgrades/figures/multivar_covariates.png	47bd6b2c4121	
results/upgrades/figures/psm_sensitivity.pdf	d196daf9b957	
results/upgrades/figures/psm_sensitivity_caliper.pdf	1077f98b8ad8	
results/upgrades/figures/psm_sensitivity_ntraw.pdf	84fd98a3aa0b	
results/upgrades/fm_cross_architecture.csv	8370f9a4a59c	14
results/upgrades/fm_finetune_v2_ccre_enrichment.csv	0547f0369d08	60
results/upgrades/fm_finetune_v2_ccre_sensitivity.csv	15e20df92a4c	42
results/upgrades/fm_finetune_v2_compare.csv	cbe379f5df0b	25
results/upgrades/fm_finetune_v2_compare_paired.csv	eaa76582835e	66
results/upgrades/fm_finetune_v2_gc_paired.csv	eb3df8dd6a9d	58
results/upgrades/fm_finetune_v2_model_contrast.csv	1b13904697d4	4
results/upgrades/fm_finetune_v2_paper_table.csv	53f2b63dd34d	5
results/upgrades/fm_finetune_v2_posthoc_paper_table.csv	1ebed5142aac	20
results/upgrades/fm_finetune_v2_randdraw.csv	04353cd3521d	25
results/upgrades/fm_finetune_v2_runs.csv	10abd8b5aca8	54
results/upgrades/fm_finetune_v2_token_lengths.csv	9a4411634f3a	9
results/upgrades/fm_finetune_v2_unit_diagnostics.csv	954584e45efb	54
results/upgrades/fm_head_invariance.csv	d3b668064a12	12
results/upgrades/gue_emp_provenance.csv	bfa9d62057dc	10
results/upgrades/gue_failure_route_audit.csv	ca9c21435e35	12
results/upgrades/gue_family_summary.csv	271ebc1d1fe3	7
results/upgrades/gue_overlap_summary.csv	adc917f5ea6d	28
results/upgrades/gue_paper15_reference.csv	c6e8b18de19d	15
results/upgrades/gue_predictions.csv	ed1683241312	7
results/upgrades/gue_published_comparison.csv	74d413e50a0a	28
results/upgrades/gue_results.csv	890be79caa86	28
results/upgrades/gue_rule_sensitivity_summary.csv	c6d8b06e5581	12
results/upgrades/gue_twins.csv	6c7349e8d6c8	28
results/upgrades/homology_recluster_audit_task.csv	df30d305cebc	30
results/upgrades/homology_recluster_e1_draws_summary.csv	c434f5045a77	3
results/upgrades/homology_recluster_extra_cleaning.csv	af66ac83c80f	63
results/upgrades/homology_recluster_extra_coords.csv	a4759c995392	3
results/upgrades/homology_recluster_extra_fm.csv	a8ef9f34df6a	420
results/upgrades/homology_recluster_extra_graded.csv	757d574bbde5	12
results/upgrades/homology_recluster_extra_specificity.csv	ac5e166e0b84	21
results/upgrades/homology_recluster_ft_contrasts.csv	f2b6c5b0ec73	45
results/upgrades/homology_recluster_ft_split_manifest.csv	0b0070fab0b6	24
results/upgrades/homology_recluster_interventions.csv	014f3011c385	36
results/upgrades/homology_recluster_labels_addendum.csv	614103680929	30
results/upgrades/homology_recluster_paper_table.csv	eedb182748de	15
results/upgrades/homology_recluster_positional.csv	5f38a38525f1	34
results/upgrades/homology_recluster_route1.csv	f42a6e77ed8e	66
results/upgrades/homology_recluster_table.csv	83d2f22a5bb7	15
results/upgrades/multivar_covariates.csv	b8b219ad8458	15
results/upgrades/multivar_nested.csv	6b1e96eb5725	18
results/upgrades/multivar_noise.csv	8fd7392117c0	2
results/upgrades/multivar_partial.csv	955c29487da8	40
results/upgrades/multivar_posthoc_distinct_datasets.csv	36f2f9606eba	14
results/upgrades/multivar_posthoc_family.csv	9659cfbc235d	16
results/upgrades/multivar_posthoc_homology_clean.csv	a3828c35466f	4
results/upgrades/multivar_posthoc_homology_models.csv	528185575ffa	8
results/upgrades/multivar_posthoc_locus_independent.csv	c6081827393a	10
results/upgrades/multivar_posthoc_slope_transfer.csv	bb95973905ce	2
results/upgrades/multivar_posthoc_task_locus_links.csv	6aed2fff1874	55
results/upgrades/multivar_preregistration.txt	163d86e92bf3	
results/upgrades/multivar_reviewer_model.csv	55c85c333f8e	8
results/upgrades/neg_provenance_bg_pooled.csv	281e5b3200df	24
results/upgrades/neg_provenance_chrom.csv	9de1ca34c5de	50

File	SHA-256 (first 12)	Rows
results/upgrades/neg_provenance_chrom_tests.csv	cdf7868d389d	12
results/upgrades/neg_provenance_chromhmm_rule_counts.csv	77ae0b970eec	10
results/upgrades/neg_provenance_compact_table.csv	5a571cc81c08	13
results/upgrades/neg_provenance_nt_by_split.csv	0c05ecf26546	135
results/upgrades/neg_provenance_nt_origin.csv	2211e44cac90	10
results/upgrades/neg_provenance_nt_wholeclass.csv	0b00d95623bb	4
results/upgrades/neg_provenance_summary.csv	ebae0954ffe1	676
results/upgrades/neg_provenance_tests.csv	ba52e5befe61	93
results/upgrades/nt_coords.csv	7d383d653358	1
results/upgrades/psm_sensitivity_between_dataset.csv	37f8b1c96859	16
results/upgrades/psm_sensitivity_caliper_sweep.csv	ef17cd96d3ef	25
results/upgrades/psm_sensitivity_decision.csv	f56f9da0f718	5
results/upgrades/psm_sensitivity_decomposition_ci.csv	21e7fe95bd0c	90
results/upgrades/psm_sensitivity_dissociation_stats.csv	9feb7dce61b4	15
results/upgrades/psm_sensitivity_leakage_check.csv	554ea22f2ad9	3
results/upgrades/psm_sensitivity_selection_bias.csv	2c4bc94e2c3a	158
results/upgrades/psm_sensitivity_summary.csv	57b7ec4b859c	932
results/upgrades/route3_specificity.csv	1670240ce2f8	7
results/upgrades/route3_threshold_invariance.csv	1eac086288cb	28
results/upgrades/shuffled_neg.csv	78bad158fe3b	6
src/upgrades/alt_model/analyze_alt_model.py	fde202b4a195	
src/upgrades/alt_model/deferred/run_ebm_regulatory_full.sh	57a98df6207b	
src/upgrades/alt_model/run_alt_model.py	3f43bf011a1c	
src/upgrades/fm_finetune/README.md	1694c4c187bb	
src/upgrades/fm_finetune/eval_ft_v2.py	bbc591073d54	
src/upgrades/fm_finetune/gpu_deferred/jobs.sh	722c56f59e21	
src/upgrades/fm_finetune/posthoc_ft_v2.py	7a89b490e7ca	
src/upgrades/fm_finetune/sensitivity_ft_v2.py	bfc18facab52	
src/upgrades/gue_suite/emp_provenance.py	89d383c1ba7c	
src/upgrades/gue_suite/rule_sensitivity.py	08fa130f6f02	
src/upgrades/gue_suite/run_gue.py	86ce3787251b	
src/upgrades/homology_recluster/README.md	bcfd83f83104	
src/upgrades/homology_recluster/deferred/run_deferred_dnabert_hc.sh	0cee26bc2517	
src/upgrades/homology_recluster/run_eval.py	5d3a6d6fea59	
src/upgrades/homology_recluster/run_extra.py	eb60bf6522e8	
src/upgrades/homology_recluster/run_labels_addendum.py	fa7147d18478	
src/upgrades/homology_recluster/run_routes.py	786c9be93a58	
src/upgrades/multivar/run_multivar.py	f4e3586f7a98	
src/upgrades/neg_provenance/README.md	191336f1ec61	
src/upgrades/neg_provenance/bg_seed_stability.py	b278e8ab7fcd	
src/upgrades/neg_provenance/nt_split_provenance.py	e9667f8614d4	
src/upgrades/neg_provenance/run_neg_provenance.py	a080ec7e56fc	
src/upgrades/psm_sensitivity/run_posthoc.py	8a7e529e355f	
src/upgrades/psm_sensitivity/run_psm_sensitivity.py	ba55596fe9f0	
src/upgrades/route3_specificity/make_fig_route3_specificity.py	f7a8ac00fbfa	